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Remote Sensing Data Assimilation With a Chained Hydrologic-Hydraulic Model for Flood Forecasting

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Key Points:

- A chained hydrologic-hydraulic modeling system integrating ISBA-CTRIP runoff with high-resolution hydraulic model TELEMAC-2D
- Advanced Ensemble Kalman Filter to reduce uncertainties in the hydraulic parameters and inflow discharge, improving flood predictions
- Assessment of three different forecasting strategies up to 36 hr show that using CTRIP during the forecast phase yield more accurate results

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Abstract Effective flood risk management requires reliable forecasts with extended lead times to enable the implementation of cost-effective and timely measures. In this study, we present a chained hydrologic-hydraulic modeling framework designed for effective near-real-time flood forecasting. The system integrates runoff predictions from a large-scale hydrologic model (ISBA-CTRIP) as inputs into high-resolution, local hydrodynamic models (TELEMAC-2D) to forecast water levels and flood extents. To improve the forecast accuracy, an Ensemble Kalman Filter is employed to reduce uncertainties in hydrological forcing and friction parameters by jointly assimilating in situ water level measurements and satellite-derived flood maps. Such a data assimilation framework operates in a real-time forecasting configuration, consisting of a reanalysis phase having an assimilation window up to the current time, followed by a forecast phase from the current time to the expected lead time. Three forecasting strategies were evaluated: (a) using CTRIP-predicted runoff for both reanalysis and forecast phases, (b) using observed discharge for reanalysis and CTRIP runoff for forecast, and (c) using observed discharge for reanalysis and keeping a constant discharge during the forecast. Results show that using observed discharge during reanalysis combined with CTRIP-predicted runoff for forecasting yields the most consistent performance. However, for all three strategies, forecast accuracy declines with longer lead times, especially when errors in the CTRIP forcings are non-stationary. This work highlights the potential for hydrologic model, despite their inherent imperfections, to serve as effective inputs for local hydraulic models, enabling near-real-time flood forecasting through the assimilation of in situ and remote sensing data.

1. Introduction

Early warning and prediction of flood events have become increasingly more important as the occurrence and intensity of floods have increased in recent decades, especially in the context of climate change (CRED, 2021). For operational flood forecasting, the challenges lie in producing reliable forecasts under the constraints of computational resources and processing time required for operational forecasting. Hydrodynamic or hydraulic models—which are fundamental to estimating water levels and flood extent—typically require observed data from stream gauge networks. However, securing this data can be challenging due to either a lack of in situ gauge data or the unreliability of measurements during intense flood events. As a result, prescribing discharge and/or water level as forcing time-series at the upstream and lateral boundary conditions (BCs) of the models can become problematic. Furthermore, constraining the hydraulic model to a liquid BC derived solely from observed forcing time-series limits its forecast lead time within the transfer time of the river network. To extend forecast lead times, it is necessary to introduce predicted inflow discharges into the hydrodynamic model, such as those simulated by large-scale hydrologic models (Hostache et al., 2018; Nguyen et al., 2023). However, due to their coarse resolution and simplified representation of physical processes (Munier & Decharme, 2022), these large-scale hydrologic models introduce uncertainty into the forcing of hydrodynamic models, leading to inaccuracies in flood forecasts. Improvements in large-scale hydrologic outputs are therefore imperative; this could be done through the assimilation of hydraulic variables into hydrodynamic models (Di Mauro et al., 2021; Grimaldi et al., 2016; Hostache et al., 2018).

Flood simulation and forecasting capabilities have been greatly improved by advances in data assimilation (DA). DA combines observations as they become available with numerical models to reduce uncertainties in the model state, parameters, and/or BCs. A classical DA approach is the assimilation of water surface elevation data, either from in situ gauge measurements, from altimetry satellites, or retrieved from remotely-sensed images (especially

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radar data) using flood edge location information combined with complementary Digital Elevation Model data (Dasgupta et al., 2021; Garambois & Montazem, 2020; Grimaldi et al., 2016). Satellite synthetic aperture radar (SAR) data are particularly advantageous for flood studies because they provide global, all-weather, day-and-night coverage of the continental water surface, which is characterized by low backscatter values resulting from the specular reflection of the incident radar pulses (Martinis et al., 2015). Numerous studies have suggested the assimilation of RS-derived water surface elevation, as reviewed by (Revilla-Romero et al., 2016). However, instead of deriving water surface elevation from flood extents, it is possible to directly assimilate SAR-derived flood probability maps (Hostache et al., 2018) or flood extent maps (Nguyen, Ricci, Piacentini, et al., 2022). DA has also proven valuable in streamflow forecasting, whether through the use of the Ensemble Kalman Filter (EnKF) (Le et al., 2022; Moradkhani et al., 2005), Particle Filter approaches (Abbaszadeh et al., 2018, 2020; García-Alén et al., 2023), variational (Garambois & Montazem, 2020), or hybrid variational-ensemble methods (Abbaszadeh et al., 2019). The reliability and accuracy of real-time forecasts are strongly influenced by the integration of observational data and predictive models using DA techniques (Foroumandi et al., 2025; Liu et al., 2012).

In the present paper, the in situ water level observations are assimilated, using an EnKF, together with 2D flood extent observations derived from Sentinel-1 SAR images, expressed as wet surface ratio (WSR) computed over the floodplain subdomains (Nguyen, Ricci, Piacentini, et al., 2022). This ratio comes directly from observed flood extent maps, which are derived from SAR images processed by a random forest algorithm from Sentinel-1 images. It is the ratio of the number of wet pixels in the floodplain subdomain to the total number of pixels in the subdomain. The proposed DA algorithm reduces the uncertainties in the hydrodynamic model parameters as well as in the forcing inputs, provided by in situ measurement at gauging stations or by large-scale hydrologic models. We propose a chained hydrologic-hydrodynamic workflow designed for real-time forecasting of hydraulic variables over a reach of the Garonne River in the southwest of France, and a reach of the Ohio River in the United States. This is orchestrated as an ensemble-based cycled DA procedure that stands in a reanalysis phase having an assimilation window up to the current time, followed by a forecast phase from the current time to the expected lead time. Different strategies using observed or simulated forcing time-series, during both the reanalysis and forecast phases, are applied to several major events on the Garonne and Ohio Rivers. Two significant floods occurred along the Garonne River—in January–February 2021 and in December 2019—hereafter referred to as Events A and B, respectively. A major flood also occurred along the Ohio River in February 2018, referred to as Event C. This study builds on our previous research works (Nguyen, Ricci, Fatras, et al., 2022, Nguyen, Ricci, Piacentini, et al., 2022, Nguyen et al., 2024, 2025), with a similar DA strategy using the same TELEMAC-2D (T2D, www.opentelemac.org) hydrodynamic model for the Garonne catchment, and additionally incorporates a dedicated T2D hydrodynamic model developed for the Ohio catchment. It further introduces a key advancement: a focus on operational forecasting—beyond reanalysis alone—with extended lead times enabled by a coupled hydrologic–hydraulic modeling system.

The remainder of the article is organized as follows. Section 2 describes the hydrologic and hydrodynamic models used in this work as well as the study areas and flood events, while Section 3 presents the DA workflow for reanalysis and forecast. Section 4 provides the results with comprehensive assessments in the control and observation spaces, for the reanalysis experiments and the forecast at increasing lead-times. Conclusion and perspectives are lastly provided in Section 5.

2. Models and Data

2.1. Hydrologic and Hydraulic Models

The general workflow for this strategy is shown in Figure 1. Here, the chained multi-physics and multi-scale modeling approach involves a direct supply of runoff forcing data to a hydraulic model with DA (purple block in Figure 1) by a hydrologic model (blue block in Figure 1), as an alternative to the discharge observed at an in situ gauging station (orange block in Figure 1). The ISBA-CTRIP hydrologic model results from coupling the ISBA (Interaction Sol-Biosphère-Atmosphère) land surface model (LSM) (Noilhan & Planton, 1989) forced by SAFRAN meteorological data for the Garonne catchment or ERA5 for the Ohio catchment, and a CNRM-modified version of the TRIP (Total Runoff Integrating Pathways) river routing model (RRM) (Oki & Sud, 1998).

SAFRAN (Durand et al., 1993; Le Moigne et al., 2020; Quintana-Segui et al., 2008) provides a high-resolution, hourly reanalysis of near-surface meteorological variables, including air temperature and specific air humidity at

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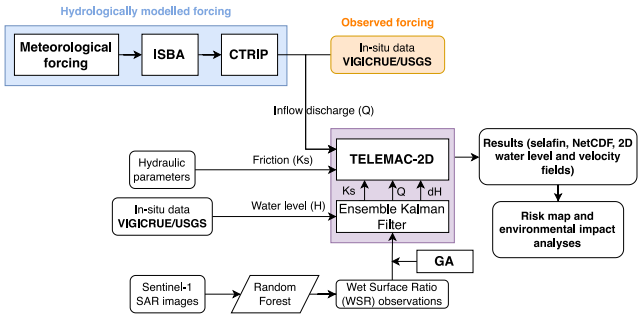
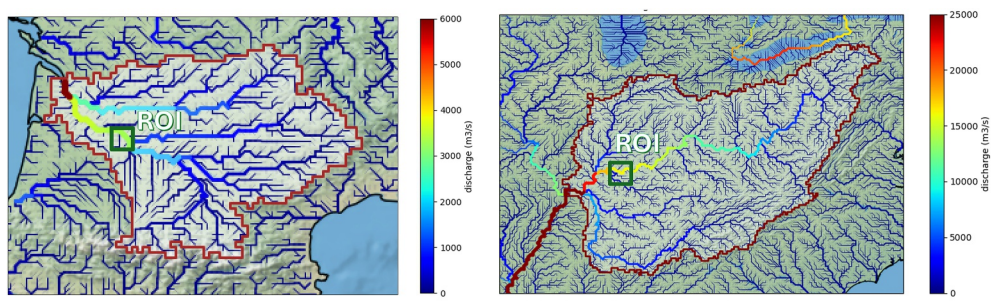


Figure 1. Workflow of the proposed data assimilation strategy for flood forecasting, built on top of the TELEMAC-2D model (purple block) to correct the hydrologic inputs and hydraulic parameters, with the inflow discharge provided by the hydrologic model (blue block), in place of the observed discharge (orange block).

2 m, wind speed at 10 m, direct and diffuse solar radiation, and rainfall rate across France. The analysis was conducted over homogeneous zones of several hundred square kilometers, with explicit consideration of the vertical dimension using 300 m elevation slices. The resulting fields are interpolated onto an 8 × 8 km^2 grid (with 9,892 cells in total), taking into account the average altitude of each grid cell. These gridded data sets, derived from local daily observations, serve as input to the ISBA LSM and support hydrometeorological applications across France and neighboring border regions. On the other hand, ERA5 (Hersbach et al., 2020) is the fifth generation of global atmospheric reanalysis produced by ECMWF, spanning from 1950 to the present, with a 31-km horizontal resolution, 137 vertical levels to 1 Pa, and hourly outputs. In this study, ERA5 provides the meteorological forcing for the ISBA-TRIP hydrologic model to simulate river discharge globally. However, since ERA5 can generally underestimates precipitation, the ERA5 precipitation data are bias-corrected using the monthly precipitation product from the Global Precipitation Climatology Center data set. This adjustment significantly enhances the agreement between ISBA-TRIP simulated discharges and observed river flows from the Global Runoff Data Center (Szczypta et al., 2012).

LSMs simulate the energy and water balance at the interface of soil, atmosphere, and vegetation, while RRMs simulate the lateral movement of freshwater toward the continent-ocean interface. The ISBA model operates globally on a 0.5° × 0.5° regular grid, calculating the energy and water budget over land surfaces with a three-layer soil representation. It provides estimates of surface runoff and gravitational drainage, which are then used as input for the CTRIP model. CTRIP, defined on a regular latitude-longitude grid with a resolution of 1/12° (Munier & Decharme, 2022), uses a river network to laterally transfer water from one grid cell to another, eventually reaching the ocean. This present work is based on the recent CTRIP version from (Decharme et al., 2012, 2019) that provides a gridded map of discharge, illustrated in Figure 2 over the Garonne (France) and Ohio (USA) catchments, respectively on 2021-02-03 (Event A) and 2018-02-28 (Event C). The CTRIP simulation time step is set to 1 hr, and the output streamflow time step is also hourly over the Garonne catchment, whereas they are both daily for the Ohio catchment. In Figure 2, the green rectangles show the regions-of-interest within the Garonne or Ohio River catchments, represented by the red polygons. The uncertainty in these simulated discharges mainly stems from uncertainties in the LSM precipitation inputs, RRM parameters and catchment description (Munier & Decharme, 2022).

River hydrodynamic or hydraulic models take into account the volume of water entering a river system (either observed by a streamgauge station or simulated by a hydrological model) to calculate water levels and velocities in the river channel, and in the floodplain once the channel's capacity is exceeded. Free-surface hydraulic modeling is governed by the Shallow Water Equations (SWE), which are derived from depth-averaged Navier-Stokes Equations. In the present work, the BC for local-scale and high-fidelity hydrodynamic models with T2D is provided by the large-scale hydrologic model ISBA-CTRIP (Decharme et al., 2019; Munier & Decharme, 2022). T2D allows to solve the SWE with an explicit first-order time-integration scheme, a finite-element method, and an



(a) ISBA-CTRIP over Garonne 2021-02-03 (b) ISBA-CTRIP over Ohio 2018-02-28

Figure 2. River discharge ($m^3.s^{-1}$) simulated by ISBA-CTRIP over the (a) Garonne catchment (Event A), and (b) Ohio catchment (Event B). The green rectangles show the regions-of-interest (ROI) in this study.

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Table 1
Study Flood Events

Study flood event	A	B	C
Catchment	Garonne	Garonne	Ohio
Drainage area [km^2]	50,100	50,100	251,228
Time period	January–February 2021	November 2019	February–March 2018
Number of mesh nodes	41,416	41,416	104,321
Max discharge at upstream BC (m^3.s^{-1})	6,190	5,760	21,747
Number of riverbed/floodplain friction coeff.	6/1	6/1	2/1
Number of floodplain subdomains	5	5	11
Number of in situ stations	3	3	2
Ensemble size (N_e)	75	75	48

iterative conjugate gradient approach (Hervouet, 2007). This allows for high precision in representing complex geometries and high fidelity in representing the hydraulic state. In addition, T2D is open-source and highly parallelizable, making it suitable for computationally demanding simulations like ensemble DA, serving to quantify and reduce the inherent uncertainties in forcing data and model parameters.

2.2. Study Areas and Flood Events

The chained hydrologic-hydraulic modeling framework for flood forecasting is carried out on two study areas, and for three flood events. Table 1 summarizes all three flood events along with the differences in the hydraulic model setup and parameters. The first study area, shown in Figure 3a, extends over a 50-km stretch of the Garonne River between Tonneins and La Réole, in the southwest of France. Gauging stations operated by EauFrance and used for the VigiCrue (https://www.vigicrues.gouv.fr) platform, located at Tonneins, Marmande and La Réole (black circles in Figure 3a), provide water level observations every 15 min. The local T2D hydraulic model over this reach was developed by Electricité de France (EDF) and presented in (Besnard & Goutal, 2011; Nguyen, Ricci, Fatras, et al., 2022). The Strickler friction coefficient (Gauckler, 1867) is assumed to be uniform over each of six segments of the riverbed K_{s_k} (with $k \in [1,6]$), indicated by solid-colored segments in Figure 3a, and also uniform over the entire floodplain (K_{s_0}). The second study area over Ohio River is shown in Figure 3b. The T2D hydraulic model for the Ohio River between the two USGS gauge stations at Cannelton and Newburgh (black circles in Figure 3b) was developed at CERFACS, in collaboration with NOAA National Weather Service (NWS) and Ohio State University (Ricci et al., 2024). For the riverbed, two uniform Strickler friction coefficients are defined for the two river segments (K_{s_k}, $k \in [1,2]$; solid-colored sections in Figure 3b), along with a single coefficient for the floodplain (K_{s_0}). Such an approach to floodplain friction is primarily based on the fact that in situ observations used for calibration are mostly available within the river channel and rarely in the floodplain. It is worth-noting that all friction coefficients K_{s_k}, are subjected to be modified over time during the DA. Furthermore, the typically low flow velocities in the floodplain lead to slower dynamics, which are in fact influenced more strongly by the topography and the numerous dikes and structures in the area. As a result, the model's sensitivity to friction changes in the floodplain is reduced, and this sensitivity is expected to decrease even more as it gets fully inundated, due to the dominant role of the continuity equation. Furthermore, the limited availability of in situ gauging observations introduces errors in the formulation of the rating curve, which is used to convert observed water levels into discharge. This is particularly problematic for high flows, as it often requires extrapolation beyond the range of typically measured values. In the present study, the discharge time-series at Tonneins for Events A and B and at Cannelton for Event C, that is, the forcing data for the T2D model, is provided either by in situ VigiCrue/USGS observations (orange block in Figure 1) or by ISBA-CTRIP simulation (blue block in Figure 1), both of which are highly prone to uncertainties but at various levels.

While observed discharge offers ground-truth data, hydrological models provide broader spatial and temporal coverage, enable forecasting, and are essential in ungauged basins or operational contexts where timeliness are crucial. In this study, we use the ISBA-CTRIP model due to its proven ability to simulate the water cycle in a spatially distributed manner across regional to global scales. Its robustness and versatility have made it a widely

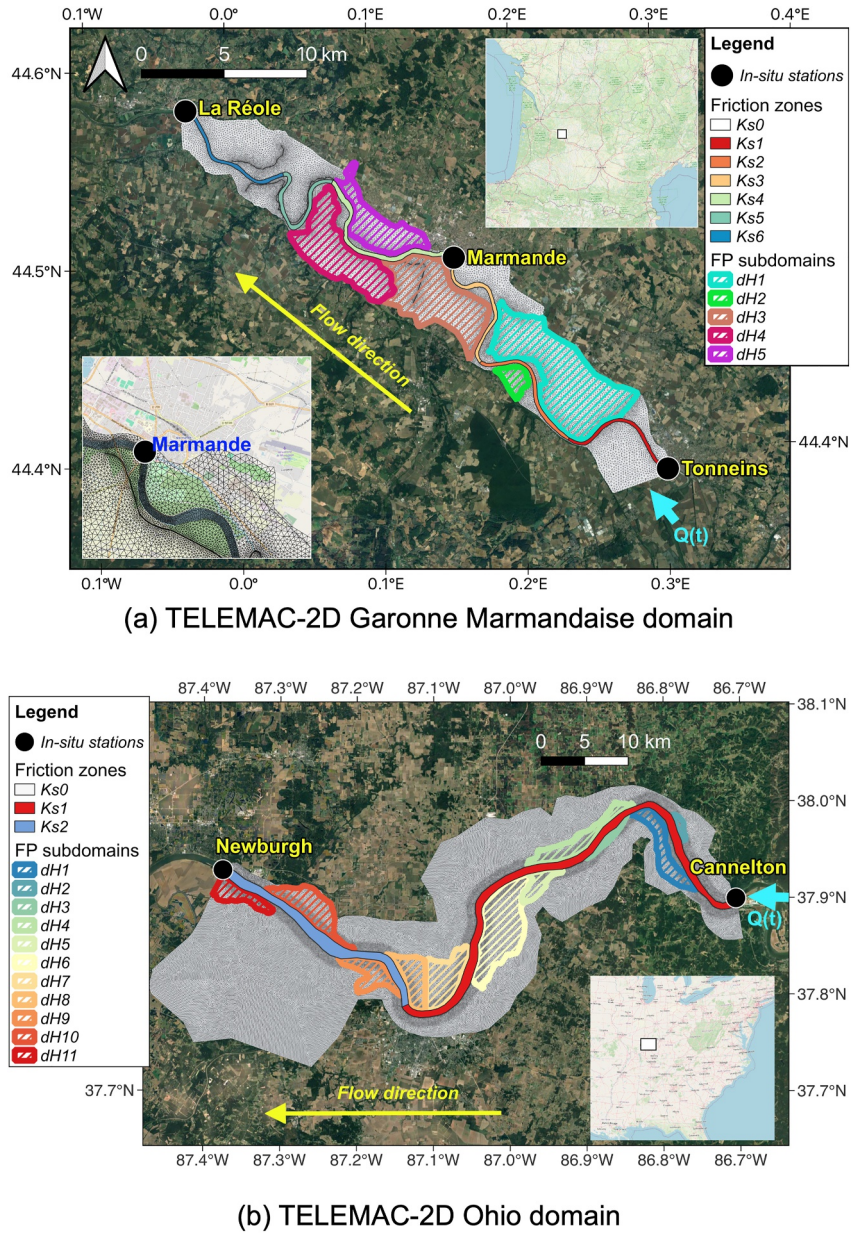


Figure 3. (a) TELEMAC-2D Garonne Marmandaise domain (from Nguyen, Ricci, Piacentini, et al. (2022)) for Events A and B, and (b) TELEMAC-2D Ohio domain for Event C. The in situ gauging stations are represented by black circles. The different river friction zones are indicated as colored segments. The floodplain is divided into subdomains that are hatched in different colors.

adopted tool in both research and operational hydrology (Decharme et al., 2019; Munier & Decharme, 2022). However, it is worth noting that the ISBA-CTRIP hydrologic model generally gives better performance for large basins, and that moderate performance can be expected for the medium-sized Garonne basin at Marmande, especially for high flows. In fact, for large flood events, ISBA-CTRIP tends to underestimate the discharge. Figures 4a and 4b show the CTRIP simulated runoff (blue) along with the observed discharge (orange) at Tonneins VigiCrue station for the flood event A and B that occurred in 2021 and 2019, respectively; the acquisition time of the Sentinel-1 SAR images is indicated by vertical dashed lines. The 2021 flood event (Event A) has an estimated return period of 20 years, while the return period of Event B (2019) is estimated between 10 and 15 years. The 2019 flood had been the most severe event in the study area since December 1981, until it was surpassed by the 2021 flood (Marchandise et al., 2024). Both events have been observed by multiple Sentinel-1 overpasses. Figure 4c show the

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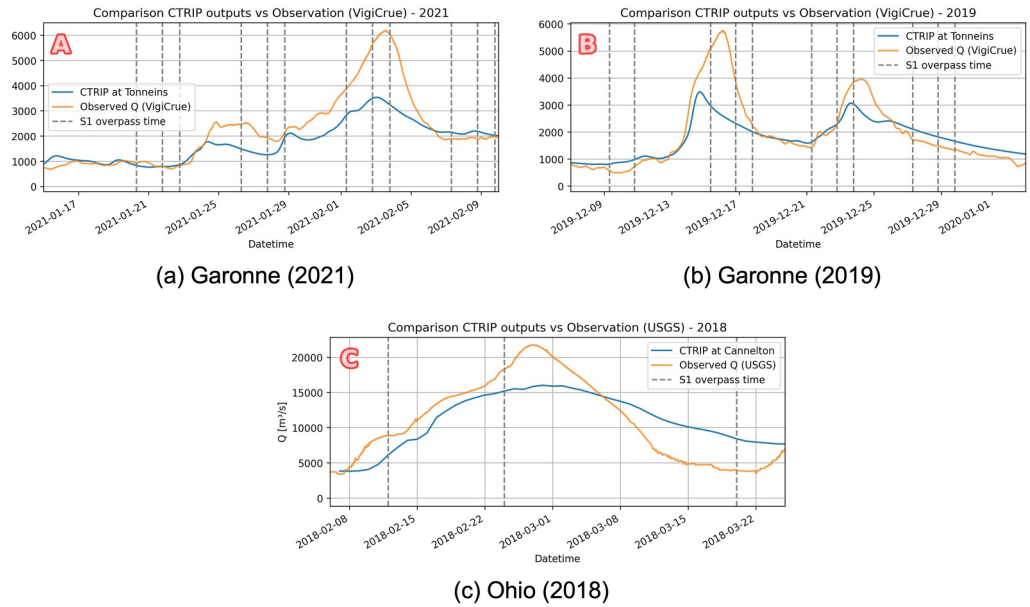


Figure 4. Discharge time-series observed at VigiCrue/USGS gauging station (orange line) and simulated by ISBA-CTRIP (blue line) at the upstream boundary condition (Tonneins/Cannelton) for Event A, B, and C. The overpass times of Sentinel-1 synthetic aperture radar are indicated with vertical dashed lines.

CTRIP simulated runoff at Cannelton USGS station along with the observed discharge, respectively in blue and orange, for the 2018 flood event (Event C). The NOAA NWS classified the February 2018 flood as a top-10 event on the Ohio River, with water levels reaching 15.45 m in Cannelton and 15.28 m in Newburgh. Due to the reduced temporal revisit frequency of Sentinel-1 over the United States compared to Europe, only three acquisitions were available during this flood event. On average, CTRIP underestimates the observed flood peak discharges by approximately 36% across the three events.

2.3. Sentinel-1 SAR-Derived Flood Observations

SAR offers a significant advantage in flood monitoring and mapping, offering the ability to capture surface water and flooded areas globally, in near real-time, and in all weather conditions. This is made possible by detecting low backscatter pixels, which result from the specular reflection of radar pulses upon their arrival at the water surfaces (Martinis et al., 2015). In the present research work, the Sentinel-1 SAR data (C-band) acquired in Interferometric Wide swath mode and Ground Range Detected (GRD) at High resolution is used. The GRD products consist of focused SAR data that has been detected, multi-looked and projected to ground range, and distributed at a 10×10 m resolution. These products are applied to generate binary water maps using a Random Forest classifier (Pal, 2005), implemented in the FloodML software (Huang et al., 2020; Kettig et al., 2021; Rodriguez Suquet et al., 2023). FloodML, which delivers flood and non-flood classification with Machine Learning methods, achieves an overall accuracy of 86.86% when validated across five test sites worldwide (Kettig et al., 2021). The code is available at https://github.com/CNES/floodml and also at https://zenodo.org/records/16565154 Rodriguez Suquet (2025), and further details can be found in (Nguyen et al., 2021, Nguyen, Ricci, Fatras, et al., 2022).

Similar to other active coherent imaging systems, SAR images are affected by an intrinsic error known as *speckle*. This occurs due to the constructive and destructive interference of coherent waves reflected from multiple individual scatterers within a single resolution cell. Typically, radiometric and geometric corrections (Blacknell et al., 1989; Small, 2011) are applied as standard pre-processing steps to raw SAR data, significantly reducing other errors, leaving speckle as the primary remaining source of error in the images. Water bodies and flooded areas generally display low backscatter (BS) in SAR images because the incident radar pulses are mostly reflected specularly away from the water surfaces. This makes the detection of inundated areas relatively straightforward, with some exceptions such as urban environments, vegetated regions, or when there is variability in water surface roughness and speckle effects. Flooded vegetation areas (e.g., partially submerged vegetation) are often misclassified due to SAR signals (e.g., in C-band and X-band) being caught in volume scattering within the canopy,

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preventing them from reaching the water surface below, or due to multiple-bounce effects between tree trunks and the water surfaces (Martinis & Rieke, 2015; Pierdicca et al., 2018). Similar challenges arise in urban areas due to complex landscape geometries, such as shadowing, layover, and highly reflective scatterers. Taking into account such imperfection and the potential non-Gaussianity of SAR-derived flood observations—that could cause suboptimality for the EnKF—a Gaussian anamorphosis (GA) process was proposed and integrated into the EnKF (Nguyen et al., 2024).

3. Data Assimilation

3.1. Cycled DA Analysis

The present research work is a continuation of (Nguyen, Ricci, Piacentini, et al., 2022, 2024). While Nguyen, Ricci, Piacentini, et al. (2022) introduced a dual state-parameter approach for assimilating Sentinel-1-derived WSR observations, Nguyen et al. (2024) later addressed their non-Gaussian characteristics using anamorphosis. However, both studies considered only the uncertainty in observed inflow discharges at Tonneins, within the reanalysis mode. In contrast, the present work focuses on the more uncertain discharges generated by a large-scale hydrologic model, that is, ISBA-CTRIP, for the sake of extending forecast lead time, within an operational forecast mode. Indeed, it is assumed that the main sources of uncertainty in the chained hydrologic-hydraulic model lie in the ISBA-CTRIP-simulated runoff, in the T2D friction parameters, as well as in the simulated hydraulic state in the floodplain. Such a DA strategy was also validated over a similar chained hydrologic-hydraulic model, but within the framework of a twin experiment with synthetical observations previously carried out in (Nguyen et al., 2023). The errors in the ISBA-CTRIP forcing data are accounted for by a multiplicative factor μ applied to the inflow discharge time-series. This correction factor is constant over a DA cycle and varies between DA cycles. In addition, the errors in the hydraulic state due to the absence of evapotranspiration and soil infiltration in the T2D hydraulic model, are accounted for as a correction to the water level δH_k (where $k \in [1,5]$ for Events A and B, and $k \in [1,11]$ for Event C) over the selected subdomains of the floodplain, represented by the hashed regions in Figure 3. Within each of these floodplain subdomains, the hydraulic state correction is applied uniformly, that is, the same value for each mesh node that belong to the same subdomain. Thus, the DA control space contains the friction coefficients, a multiplicative forcing factor, and the uniform hydraulic state corrections in the floodplain. The statistical distribution for each element of the control vector is assumed to be Gaussian, centered at the a priori value (calibrated values for friction coefficients, 1 for the discharge multiplicative coefficient μ, and 0 for additive correction to the hydraulic state δH). Further details are given in (Nguyen et al., 2024; Table 1). For Events A and B, the EnKF involves 75 members with perturbed control elements, whereas the experiments for Event C include only 48 members to reduce computational cost. This ensemble size was selected to maintain computational efficiency while providing a sufficient size for the error covariance representation.

The EnKF algorithm is preferred in this study because it enables stochastic estimation of the covariance matrices between model inputs/parameters and outputs, without requiring the formulation of the hydrodynamic model's tangent linear. In addition, the uncertainty reduction due to a joint state-parameter sequential correction by an EnKF is also enhanced by a GA (Simon & Bertino, 2009, 2012). This GA algorithm allows us to deal with non-Gaussian errors associated with WSR observations by mapping these non-Gaussian measurements onto a Gaussian space (Nguyen et al., 2024). This favors the preservation of the optimality of the performed EnKF. The water level time-series at in situ observing stations and the WSR computed over the floodplain subdomains are jointly assimilated with the EnKF algorithm. This allows a sequential correction of the friction, the inflow discharge and the water level in the floodplain subdomains.

The cycled DA workflow is illustrated in Figure 5. It is implemented for a real-time forecasting system with DA cycles of 18 hr with 6 hr of overlap. Each cycle c consists of a reanalysis phase (green blocks) and a forecast phase (red blocks). The forecast phase may or may not be activated for a cycle; it has no impact on the cycled analysis. The analysis at cycle c assimilates available observations over an 18-hr window up to the present time $T_{0,c}$. This window length was selected to fully take advantage of the RS data that are sparse in time. The resulting corrected friction, forcing and corrective terms to the hydraulic state obtained from the 18-hr analysis window at cycle c are used to run a 6-hr updated simulation (green blocks, between $[T_{0,c} - 18; T_{0,c} - 12]$) that represents the hindcast reanalysis of the event. The next DA cycle $c + 1$ starts from the end of the 6-hr reanalysis at $T_{0,c+1} - 18$ hr (i.e., $T_{0,c} - 12$ hr).

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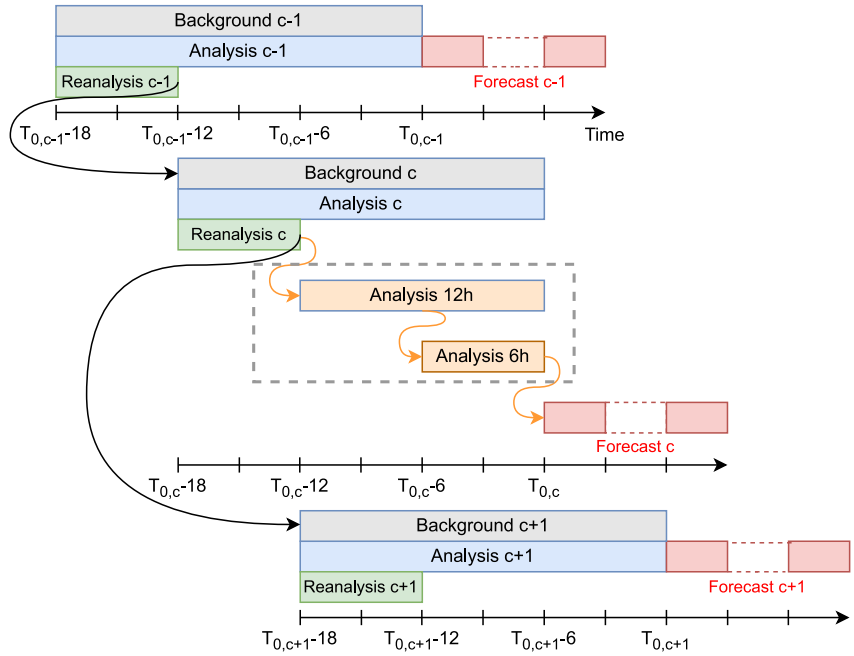


Figure 5. Data assimilation workflow for reanalysis (green blocks) and forecast (red blocks) at cycle $c - 1$, c, and $c + 1$.

3.2. DA-Based Forecast

The forecast framework is embedded in the cycled DA procedure. When activated for cycle c (e.g., during a rising limb to predict the possible upcoming flood), it articulates as three nested assimilations over c, first of which is the previously mentioned analysis (blue block in Figure 5). Taking into account only observations up to the present $T_{0,c}$, and in order to benefit from sparse RS observations while being consistent with the most recent observations, the 18-hr analysis is followed by 12-hr and 6-hr nested analyses. These two subsequent assimilations only serve for issuing the forecast from cycle c. They are represented as orange blocks in Figure 5. Each orange block stands in a background step and an analysis step as previously described. These forecast-dedicated steps are framed in the gray-dashed rectangle in Figure 5 (shown for cycle c only). They act as external loops (gradually taking into account non-linearity in the filter) and provide the optimal initial state and parameters for the forecast simulation starting at $T_{0,c}$ and lasting for 36 hr. Details on the computational cost of the DA workflow are provided in Appendix. The forcing data to the T2D hydraulic model is either supplied by a hydrologic model, or using the direct discharge observed from the Tonneins VigiCrue in situ station (for Events A and B), or from the Cannelton USGS in situ station (for Event C). For consistency, the observed discharge for Event C is also referred to as VigiCrue in the name of the strategies. Using these two different forcing data, three strategies for forcing are investigated here:

- Using CTRIP predicted runoff for analysis and forecast, called CTRIP-CTRIP (or CC) in the following,
- Using observed VigiCrue discharge for analysis and CTRIP for forecast, called VigiCrue-CTRIP (or VC) in the following, and
- Using observed VigiCrue discharge for analysis with a persistent value for forecast (i.e., keeping the last available observed value), called VigiCrue-constant Q (or VQ) in the following.

The VQ strategy, which maintains a constant observed discharge throughout the forecast period, reflects a common practice in operational flood monitoring (typically with an OL run without DA), and in that sense be considered as a baseline. All these strategies for the forecast experiments are gathered in Table 2. It should be noted that in the following, the ensemble and its mean state are used to evaluate the performance of the DA method in forecast mode.

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Table 2
Acronym of the Experiments With Different Forcing Strategies in Forecast

		Forecast	
		CTRIP	Constant Q
Reanalysis	CTRIP	CC	–
	VigiCrue	VC	VQ

3.3. Description of the EnKF Algorithm

In the following, the index of the member within an ensemble of size N_e is denoted by $i \in [1, N_e]$. The vectors $\mathbf{x}_c^{b,i}$ and $\mathbf{x}_c^{a,i}$ represent, respectively, the background and analysis control vector for member $i \in [1, N_e]$ over a DA cycle c. The EnKF background step (represented by gray blocks in Figure 5) consists in propagating the N_e control and model state vectors (i.e., $\mathbf{x}_c^{b,i}$ and $\mathbf{s}_c^{b,i}$, respectively) in time over $[T_{0,c} - 18\text{ hr};\ T_{0,c}]$ with $T_{0,c}$ being the present time for cycle c. Then, the EnKF analysis step updates the control $\mathbf{x}_c^{a,i}$ and computes the associated hydraulic state vector $\mathbf{s}_c^{a,i}$ over $[T_{0,c} - 18\text{ hr};\ T_{0,c}]$ (blue blocks in Figure 5). The first 6 hr of this integration (green block over $[T_{0,c} - 18\text{ hr};\ T_{0,c} - 12\text{ hr}]$) is the contribution from cycle c to the reanalysis for the simulated event in hindcast mode.

In the EnKF background step, the background hydraulic state for each member $\mathbf{s}_c^{b,i}$ is obtained by the integration of the hydrodynamic model $\mathcal{M}_c: \mathbb{R}^n \rightarrow \mathbb{R}^m$ which maps the control space (of dimension n) to the model state (of dimension m) over cycle c:

$$\mathbf{s}_c^{b,i} = \mathcal{M}_c\left(\mathbf{s}_{c-1}^{a,i}, \mathbf{x}_c^{b,i}\right), \tag{1}$$

where $\mathbf{s}_{c-1}^{a,i}$ stands for the hydraulic state from the previous analysis, and $\mathbf{x}_c^{b,i}$ is the background control vector at the current cycle c. To prevent ensemble collapse, artificial dispersion is introduced by adding perturbations $\boldsymbol{\theta}_c^i$ to the mean analysis from the previous cycle, such that $\mathbf{x}_c^{b,i} = \overline{\mathbf{x}_{c-1}^a} + \boldsymbol{\theta}_c^i$. These perturbations are sampled from a Gaussian distribution with zero mean and a standard deviation derived from a linear combination of the previous cycle's analysis spread and the default parameter-specific standard deviation. Full details can be found in (Nguyen, Ricci, Fatras, et al., 2022). It should be noted that, in order to avoid inconsistencies between the states and the analyzed set of parameters, a 3-hr spin-up integration preceding each 6-hr analysis window was carried out.

The observation vector, denoted as $\mathbf{y}_c^o$, contains in situ water level measurements and S1-derived WSR observations for cycle c. In a stochastic EnKF implementation (Asch et al., 2016), the observation vector $\mathbf{y}_c^o$ is perturbed by adding an observation error term $\boldsymbol{\epsilon}_c$, generating an ensemble of observations $\mathbf{y}_c^{o,i}$. The observation error $\boldsymbol{\epsilon}_c \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_c)$, with the observation error covariance matrix denoted by $\mathbf{R}_c = \sigma_{\text{obs}}^2 \mathbf{I}_{n_{\text{obs}}}$. $\mathbf{R}_c$ is assumed to be diagonal of standard deviation σ_{obs} as the observation errors are assumed to be uncorrelated Gaussian. It should be noted that within a cycle, the observation errors decreases as the observation time gets closer to the present time so that the recent observations weight more in the analysis than the older ones (i.e., recent observations are more reliable). For the DA experiments, the proportionality coefficient used to specify the observation error τ for in situ data is fixed to 15%, meaning that σ_{obs} amounts to 15% of the observation value, whereas the value of τ for WSR data varies from 10% to 20% depending on how early the observation time is within the 18-hr assimilation window. The equivalent of the model in the observation space, denoted by $\mathbf{y}_c^{b,i}$, is computed with the observation operator $\mathcal{H}_c: \mathbb{R}^m \rightarrow \mathbb{R}^{n_{\text{obs}}}$:

$$\mathbf{y}_c^{b,i} = \mathcal{H}_c\left(\mathbf{s}_c^{b,i}\right). \tag{2}$$

Next, the EnKF analysis step updates the control $\mathbf{x}_c^{a,i}$ and the associated model state vector $\mathbf{s}_c^{a,i}$ in an anamorphosed space (Nguyen et al., 2024) using the transformed observation operator $\tilde{\mathcal{H}}_c$:

$$\mathbf{x}_c^{a,i} = \mathbf{x}_c^{b,i} + \mathbf{K}_c\left(\tilde{\mathbf{y}}_c^{o,i} - \tilde{\mathbf{y}}_c^{b,i}\right). \tag{3}$$

The Kalman gain is further computed from covariance matrices that are stochastically estimated within the ensemble, considering anamorphosed observation vectors $\tilde{\mathbf{y}}^{b,i}$:

$$\mathbf{K}_c = \mathbf{P}_c^{\mathbf{x},\tilde{\mathbf{y}}}\left[\mathbf{P}_c^{\tilde{\mathbf{y}},\tilde{\mathbf{y}}} + \mathbf{R}_c\right]^{-1}. \tag{4}$$

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Similar to Equation 1, the hydraulic state $\mathbf{s}_c^{a,i}$ for each analyzed control vector $\mathbf{x}_c^{a,i}$ is obtained by the integration of the hydrodynamic model $\mathcal{M}_c$ using the updated parameters in $\mathbf{x}^{a,i}c$:

$$\mathbf{s}_c^{a,i} = \mathcal{M}_c\left(\mathbf{s}_{c-1}^{a,i}, \mathbf{x}_c^{a,i}\right). \tag{5}$$

Here, $\mathbf{x}^{a,i}c$ includes $(K_{s_k})_c^{a,i}, \mu_c^{a,i}$, and the state correction $\delta H_k^{a,i}$ in the floodplain subdomains over W_c, all starting from the same initial condition as the corresponding background simulation in the ensemble. It is important to note that the ensemble mean $\overline{\delta H_k^a}$ is used in place of individual $\delta H_k^{a,i}$ to maintain a smooth water level field.

When activated, the forecast step unfolds as follows. For each member, the first (respectively, second) forecast-dedicated analysis, in gray-dashed rectangle, uses the previous 18-hr (respectively, 12-hr) analyzed control vector as background from which a new shortened background trajectory is issued over $[T_{0,c} - 12\text{ hr}; T_{0,c}]$ (respectively, over $[T_{0,c} - 6\text{ hr}; T_{0,c}]$). Note that these shortened background simulations are not represented in Figure 5 for clarity purposes. Each of them is followed by an analysis (orange blocks) following Equation 3. The control vector resulting from the last 6-hr analysis, denoted $\mathbf{x}_{c,fT_0}^{a,i}$, are then used to integrate an ensemble of forecast simulations starting from the ensemble of hydraulic states at $T_{0,c}$, noted $\mathbf{s}_{c,fT_0}^{a,i}$, over 36 hr (red blocks in Figure 5):

$$\mathbf{s}_c^{f,i} = \mathcal{M}_c\left(\mathbf{s}_{c,fT_0}^{a,i}, \mathbf{x}_{c,fT_0}^{a,i}\right). \tag{6}$$

The ensemble of forecast trajectories $\mathbf{s}_c^{f,i}$ for $i \in [1, N_e]$ is further compared to the reanalysis (green blocks) and observations in order to assess the predictive capabilities of the DA workflow in real-time.

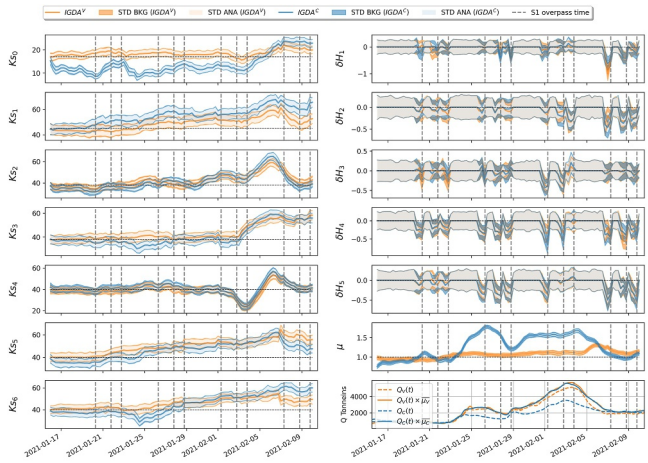
4. Results

In reanalysis phase, three types of experiment are carried out: an open-loop (OL) run without assimilation, a DA experiment (named IDA) that assimilates only in situ water-level, and a DA experiment (named IGDA) that assimilates both in situ water-level and remote-sensing WSR observations. All three flood events are simulated in hindcast mode, using either observed forcing at Tonneins/Cannelton from VigiCrue/USGS (experiments denoted by OL^{V}, IDA^{V} and IGDA^{V}), or ISBA-CTRIP forcing (experiments denoted OL^{C}, IDA^{C}, and IGDA^{C}). Then, the forecast mode is carried out from IGDA re-analysis as shown in Figure 5 using one of the three forcing strategies presented in Table 2.

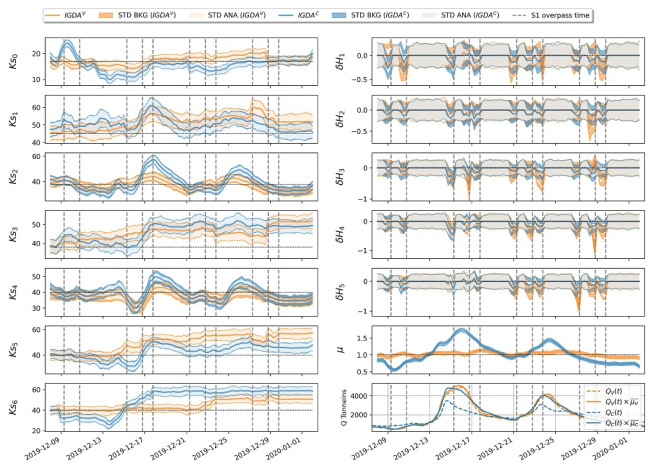
Reanalysis results are first assessed in the control space (subsection 4.1). The reanalysis and forecast results are then assessed in the observation space using 1D and 2D metrics (respectively, in subsection 4.2 and 4.3). In 1D assessment, the RMSE between observed and simulated water level at in situ observing stations is computed to assess the dynamics of the flow in the river channel. Additionally, probabilistic metrics such as the Continuous Ranked Probability Score (CRPS) (Ferro, 2014; Hersbach, 2000) and the 95% Exceedance Ratio (ER_{95}) (Moradkhani et al., 2006) are used to assess the performance of the IDA and IGDA experiments. The ER_{95} metric assesses ensemble spread by measuring how frequent the observations fall outside the ensemble-derived 95% confidence interval. Ideally, ER_{95} values should be close to 5%, indicating that observations lie outside the predictive bounds only 5% of the time. The CRPS is computed by comparing the cumulative distribution functions of the forecasts with those of a reference data set. Lower RMSE and CRPS values indicate higher forecast accuracy. On the other hand, in the 2D assessment, the agreements between flood extents simulated with T2D and flood extents derived from Sentinel-1 images are shown with a contingency map and an overall Critical Success Index (CSI) to assess the dynamics of the flow in the floodplains. The reanalysis and forecast results are finally compared to an independent High Water Marks (HWM) data set.

4.1. Reanalysis Results in the Control Space

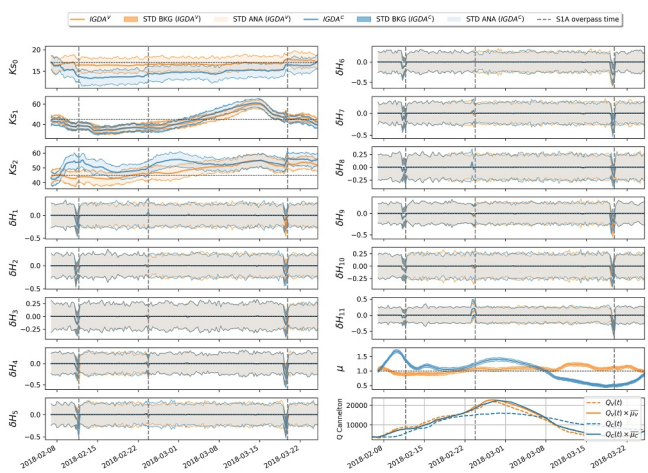
Figure 6 reveals the analyzed parameters from the different DA reanalyzes forced by VigiCrue and CTRIP for the three events, Events A (Figure 6a), B (Figure 6b), and C (Figure 6c). The solid orange and blue lines respectively represent the mean of the control vector analysis for IGDA^{V} and IGDA^{C}. The shaded envelopes represent the standard deviation of the ensemble (solid color for background and light color for analysis). They appear in a gray



(a) Event A - Garonne (2021)



(b) Event B - Garonne (2019)



(c) Event C - Ohio (2018)

Figure 6. Data assimilation analysis (IGDA) for Event A, B, and C, in the control space for friction coefficient K_{s_0} in the floodplain (left column, top panel), K_{s_k} in the riverbed, uniform water level correction δH_k, multiplicative coefficient μ for inflow discharge at Tonneins/Cannelton, and reconstructed inflow discharges.

color when all four envelopes overlap. The default values are indicated by horizontal dashed lines. The analyzed values for the friction coefficients K_{s_k} (with $k \in [0,6]$ for Events A and B, or $k \in [0,2]$ for Event C) are shown on the left column. The inflow multiplicative correction μ is shown on the right column along with the analyzed values of the hydraulic state correction δH_k (with $k \in [1,5]$ for Events A and B, or $k \in [1,11]$ for Event C). The bottom right panel displays the reconstructed upstream discharges using the mean of the μ analysis for the ensemble. The overpass times of Sentinel-1 over the flood events are indicated by vertical dashed lines.

First and foremost, the DA analysis for IGDA^{V} and IGDA^{C} provides quite similar corrections to the friction coefficients of the riverbed. Yet, the analyses for the friction in the floodplain K_{s_0} and the correction to the inflow at Tonneins (respectively, Cannelton) are very different. Indeed, CTRIP-simulated runoff is significantly under-predicted compared to the observed discharge and leads to low water levels in the whole catchment without DA (as shown below in Figure 8). As a consequence, the EnKF prescribes a multiplicative factor $\mu > 1$ to account for this lack of runoff, especially during the high flow (i.e. between 2021-01-23 and 2021-02-06 for Event A). During the flood recess, CTRIP slightly exceeds the VigiCrue observations. In order to empty the floodplain, the EnKF may either prescribe a multiplicative factor $\mu < 1$, or, for some equifinality reasons, take this overestimation into account by correcting the friction coefficients K_{s_0}, K_{s_k}, or the hydraulic state correction δH_k in the floodplain subdomains.

For all three events, since the flow velocity in the floodplain is small, the friction coefficient in the floodplain (K_{s_0}) only has a minor impact on the dynamics of the flow. This leads to small correction of K_{s_0} for IGDA^{V}. Yet, when the misfit to the observations is large (i.e., when T2D is forced with the underestimated CTRIP runoff in IGDA^{C}), the correction to K_{s_0} is large. K_{s_0} is reduced below the default value, thus increasing the water levels. Reversely, for the significantly overflowing event A, the friction in the floodplain is reduced with higher K_{s_0} during recess, to decrease the water level in the floodplain that T2D struggles to empty after the flood peak due to the absence of ground infiltration and evapotranspiration processes in the model, as well as the lack of representation of drainage systems (e.g., pumps, flood gates, or drainage network not modeled as a sufficient mesh resolution) in the hydraulic model.

The hydraulic state corrections δH_k (with $k \in [1,5]$ for Events A and B, or $k \in [1,11]$ for Event C) are visible with non-zero values at each S1 overpass. The corrections are mostly negative acting for the evacuation of water after the flood peaks. The reconstructed hydrographs at Tonneins and Cannelton for IGDA^{C}, are close to that of IGDA^{V} which illustrates that the EnKF succeeds in retrieving realistic forcing time-series even when the a priori runoff from CTRIP is significantly underestimated.

4.2. Reanalysis Results in the Observation Space

4.2.1. 1D Metrics on Water Levels at Observing Stations

Figure 7 shows the water levels at the observing stations at Tonneins (red), Marmande (blue) and La Réole (green) during Event A simulated by OL^{V} (top panel) and IGDA^{V} (bottom panel) compared to the observed water levels

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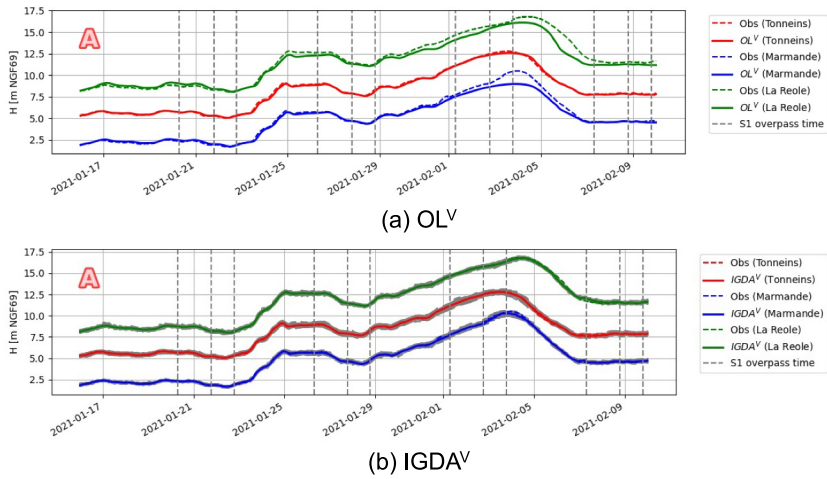


Figure 7. Water levels at the observing stations at Tonneins (red), Marmande (blue) and La Réole (green) during Event A simulated by (a) OL^V and (b) $IGDA^V$, compared to the observed water levels (plotted in dashed lines), for the experiments forced by VigiCrue discharge.

(dashed lines), for experiments forced by VigiCrue time-series. For $IGDA^V$, the ensemble of analysis is plotted in gray and the mean of the analysis is plotted in color. It should be noted that OL^V slightly underestimates the flood peak at Marmande and La Réole and that this discrepancies are efficiently reduced in $IGDA^V$ (similar results - not shown - are obtained with IDA^V). These results shows the efficiency of the DA strategy to correct the flow dynamics in the riverbed when forced by observed runoff.

Figure 8 is similar to Figure 7 but with CTRIP runoff. The underestimation of the peak is all the more visible for the simulation forced by CTRIP than by VigiCrue. Underestimated CTRIP runoff time-series leads to a significant underestimation of water level at high-flow period, including the flood peak in OL^C. DA (IDA^C and $IDGA^C$) lead to significant correction on the control leading to improved water levels at all three observing stations over the entire 2021 flood event. This illustrates how DA improves water level simulation in the river, even when the input forcing is only a coarse approximation of the real inflow, here provided by CTRIP.

Table 3 summarizes the 1D quantitative results for all six experiments, during each of the three events, at the in situ observing stations. The RMSEs between the simulated and the observed water levels at each observing station

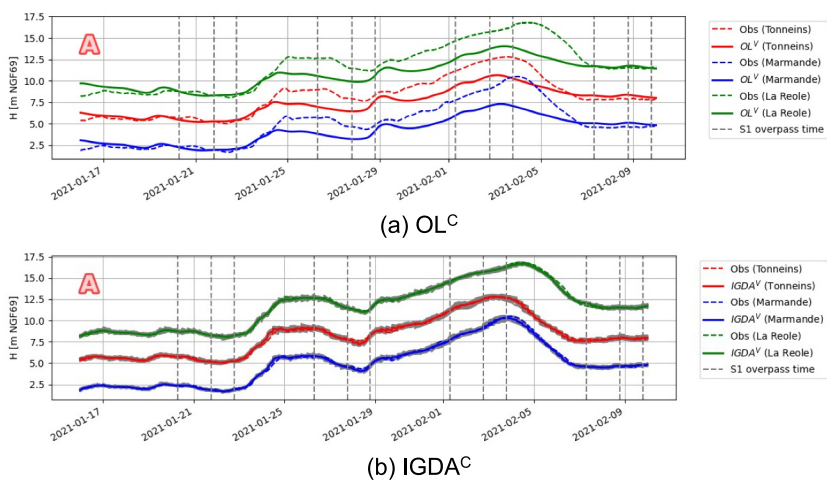


Figure 8. Water levels at the observing stations at Tonneins (red), Marmande (blue) and La Réole (green) during Event A simulated by (a) OL^C and (b) $IGDA^C$ (bottom panel), compared to the observed water levels (plotted in dashed lines), for the experiments forced by CTRIP runoff.

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Table 3
Water Level RMSE, ER_{95}, and Continuous Ranked Probability Score Computing at Observing Stations With Respect to In Situ Observed Water Levels

	RMSE [m]					
	Tonneins	Marmande	La Réole	Mean gain (%)	ER_{95} [%]	CRPS [m]
(a) Event A–Garonne (2021)						
OL^V	0.106	0.392	0.536	–	–	–
IDA^V	0.062	0.071	0.081	69.43	4.48	0.040
$IGDA^V$	0.073	0.074	0.090	65.15	5.60	0.044
OL^C	1.209	1.405	1.598	–	–	–
IDA^C	0.160	0.148	0.130	89.37	13.15	0.077
$IGDA^C$	0.166	0.160	0.141	88.69	17.47	0.083
(b) Event B–Garonne (2019)						
OL^V	0.129	0.220	0.318	–	–	–
IDA^V	0.060	0.045	0.125	64.58	2.63	0.045
$IGDA^V$	0.065	0.055	0.140	60.20	3.34	0.050
OL^C	1.109	1.224	1.375	–	–	–
IDA^C	0.208	0.181	0.257	82.59	24.98	0.128
$IGDA^C$	0.207	0.183	0.253	82.66	23.87	0.127
	RMSE [m]					
	Cannelton	Newburgh	Mean gain (%)		ER_{95} [%]	CRPS [m]
(c) Event C–Ohio (2018)						
OL^V	0.509	0.544	–		–	–
IDA^V	0.054	0.041	90.33		0.127	0.052
$IGDA^V$	0.058	0.044	90.13		0.114	0.052
OL^C	2.383	2.139	–		–	–
IDA^C	0.087	0.085	96.19		0.643	0.062
$IGDA^C$	0.093	0.082	96.13		0.586	0.062

are computed over time for Event A (Table 3a), Event B (Table 3b) and Event C (Table 3c), respectively. The probabilistic metrics ER_{95} and CRPS are computed by aggregating all stations per catchment, to provide overall assessments.

Focusing on Event A, coherent to the water level plots displayed above (Figures 7 and 8), the RMSEs underline the underestimation problem when CTRIP runoff is used without DA, resulting in errors greater than 1 m, which are critical regarding flood hazard assessments. Moreover, for both VigiCrue and CTRIP runoff, the assimilation of in situ observations only in IDA as well as the joint assimilation of in situ water level and RS-derived WSR observations in IGDA leads to a significant improvement with reduced RMSE values. Indeed, the gain with respect to OL amounts to 65.15% when using observed discharge and 88.69% when using CTRIP runoff for $IGDA^C$, with very similar results for IDA^C. This demonstrates that the assimilation of in situ data only, efficiently constrains the assimilation algorithm to properly represent the water level at observing stations. The RMSEs at observing stations from the CTRIP-forced experiment $IGDA^C$ remain slightly larger (14–17 cm) than those of the simulations forced by VigiCrue discharges (7–9 cm), yet the improvement is greater than when using VigiCrue forcing. Among the four DA experiments, the ER_{95} values indicate that both IDA^V and $IGDA^V$ closely approach the target value of 5%, while IDA^C and $IGDA^C$ also perform reasonably well, albeit with higher exceedance rates. The CRPS metric further corroborates these assessments, aligning with the RMSE and ER_{95} results and emphasizing the effectiveness of the DA strategies. Evaluation metrics for Event B and C lead to similar

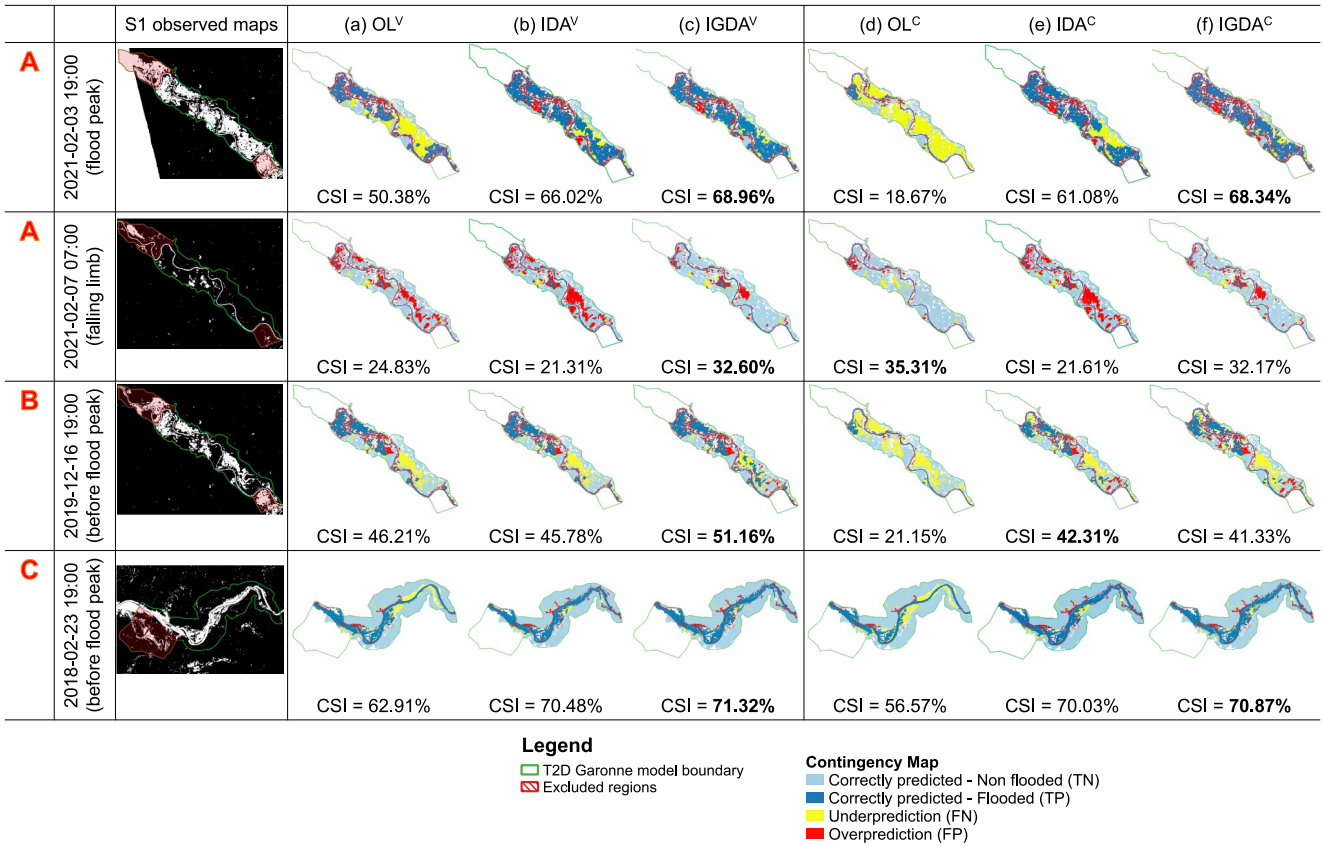


Figure 9. Contingency maps and Critical Success Index computed between simulated flood extents and S1-derived observed flood extents for (a–f) OL^V, IDA^V, $IGDA^V$, OL^C, IDA^C and $IGDA^C$, respectively, at the flood peak (first row) and at recess time (second row) for the Event A, before flood peak for the Event B (third row) and Event C (fourth row).

conclusions. Quantitative results for Event B (Table 3b) are close to those of Event A, and even better improvements are obtained for Event C (Table 3c)—whether using observed discharge or CTRIP runoff—with average performance gains exceeding 90%.

4.2.2. 2D Metrics on Flood Extent Over the Simulation Domain, Using Contingency Maps and Critical Success Index

The merits of assimilating WSR in addition to in situ data, is significant in the floodplain. The contingency maps between the simulated flood extent and the observed flood extent derived from Sentinel-1 images are computed and displayed in Figure 9. It indicates for each pixel within the studied catchment if the simulation succeeds or fails with respect to the remote sensing observation. If a pixel is correctly predicted as flooded or correctly identified as non-flooded according to the Sentinel-1-derived flood extent maps, the outcome is labeled as True Positive (TP) or True Negative (TN), respectively. Otherwise, it is labeled as False Positive (FP) or False Negative (FN), whether a pixel is not correctly predicted as flooded or not correctly predicted as non-flooded, respectively. The CSI, also called Threat Score, computes the ratio between the number of TP over the sum pixels of TP, FP, and FN. A CSI close to 100% indicates a perfect result with respect to the reference (here S1-derived flood extents).

At the flood peak on 2021-02-03 (Event A; first row of Figure 9), both OL^V and OL^C experiments show large areas of underpredicted flooding (shown by yellow areas), with OL^C being the more severe case due to CTRIP providing understimated runoff. These underprediction areas are significantly reduced by DA for both $IGDA^V$ and $IGDA^C$. It should be noted that the assimilation of only in situ data in IDA^V and IDA^C brings some improvement with respect to OL but significant underprediction remains (upstream of Marmande), especially when CTRIP runoff is used. This demonstrates the merits of the assimilation of WSR and the associated

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correction of the hydraulic state in the subdomains of the floodplain with respect to the in situ only IDA strategy. Despite the understimated CTRIP runoff, the CSI score resulting from $IGDA^C$ reaches that of $IGDA^V$. As such, the assimilation of WSR data in IGDA significantly improves the CSI score for both forcing experiments, using either CTRIP or VigiCrue forcing, with CSI of 68%–69%.

During the flood recess on 2021-02-07 (Event A; second row of Figure 9), both OL^V and OL^C experiments tend to over-predict flooding (shown by red areas) as the T2D hydrodynamic model struggles to empty the floodplain after the flood peak. The assimilation of in situ data in IDA tends to even degrade the results in the floodplain. Thus, the merits of assimilating S1-derived WSR become all the more noticeable in IGDA experiments. They significantly reduce the overflooded areas, thanks to the correction of the water level in floodplain subdomains δH_k with $k \in [1,5]$. Thanks to such a reduction, the CSI score resulting from $IGDA^V$ is improved, compared to IDA^V. When using CTRIP runoff, it can be noted that the CSI of $IGDA^C$ (32.17%) during the flood recession remains lower than that of OL^C (35.31%). Indeed, the underestimation of inflow discharge in OL^C results in the water extent remaining small at flood recess, thus yielding a slightly better CSI. In other words, the error in CTRIP's forcing acts in its favor by compensating for T2D's inability to drain the floodplain. Such an assessment (between OL^C and $IGDA^C$) should be made alongside with the 1D assessment for the same experiments (in which, the RMSE of $IGDA^C$ is reduced by 73.69% compared to that of OL^C). The fact that the CSIs from both IGDA experiments do not exceed 32% (i.e., $IGDA^V$ and $IGDA^C$ with 32.60% and 32.17%, respectively) indicates a limitation of the hydraulic model, during the flood recess. Further improvement of the model in terms of infiltration and evapotranspiration physical processes as well as in terms of drainage network representation (weirs, culverts, flap gates and ditches) should be investigated.

Similarly, significant underprediction areas by OL^V and OL^C are reduced by DA before the flood peak on 2019-12-16 (Event B; third row of Figure 9). Consistently, before the flood peak on 2018-02-23 (Event C; fourth row of Figure 9), both OL experiments exhibit extensive areas of underpredicted flooding over the eastern part of the domain. These are substantially reduced by DA. It should to be noted that for Event C, $IGDA^V$ and $IGDA^C$ experiments, lead to CSI values exceeding 70%. This result is notable, as the Sentinel-1 acquisition occurred several days before the actual flood peak on 2018-02-27 at Cannelton.

To conclude, for the three flood events, the assimilation of S1-derived WSR in reanalysis mode, both prior to and at the flood peak, produces comparable improvements with either CTRIP or VigiCrue forcing. This highlights the effectiveness of the proposed DA strategy and validates the use of CTRIP forcing despite its tendency to underestimate discharge. During recess, when CTRIP forcing is used, jointly assimilating WSR with in situ data provides some improvement compared with assimilating in situ data alone. However, no improvement over the OL is observed, as errors from CTRIP and T2D happen to compensate each other by drying out the floodplain.

4.3. Forecast Results in the Observation Space

The following subsection demonstrates the operational forecasting capability of the chained hydrologic–hydraulic modeling system with DA. In that perspective, the key results are illustrated in the observation space. The quality of the upstream forcing becomes particularly important when the lead time exceeds the river network's transfer time (typically 12 hr for Events A and B). Consequently, degraded forecasts can be expected beyond this period.

The forecasting capability of the DA strategies is evaluated considering increasing lead times, over 6 days around the flood peak during Events A and B at Tonneins, Marmande, and La Réole, and 8 days during Event C at Cannelton and Newburgh. The forecasted water levels compared to in situ observations are shown in Figure 10 for Marmande (first and second rows for Events A and B, respectively) and Cannelton (third row for Event C). Each column represents a different reanalysis/forecast forcing scenario from Table 2. In the top panel of each plot, the solid-colored lines represent forecast water levels at increasing lead times (+0 hr in blue, +6 hr in orange, +12 hr in green, +18 hr in red, +24 hr in purple, +30 hr in brown, and +36 hr in magenta). The observed water levels are shown by the black dashed line, and the blue dash-dotted line indicates OL^C simulation. The corresponding errors between forecast and observation are displayed in the bottom panels using the same color code. On each panel, the first gray vertical dashed line marks the moment when the flow transitions from a minor to a moderate flood risk, corresponding to the shift from the yellow to the orange alert level, which signals significant overflow with the potential to affect communities and safety. The second gray vertical dashed line indicate the flood peak date.

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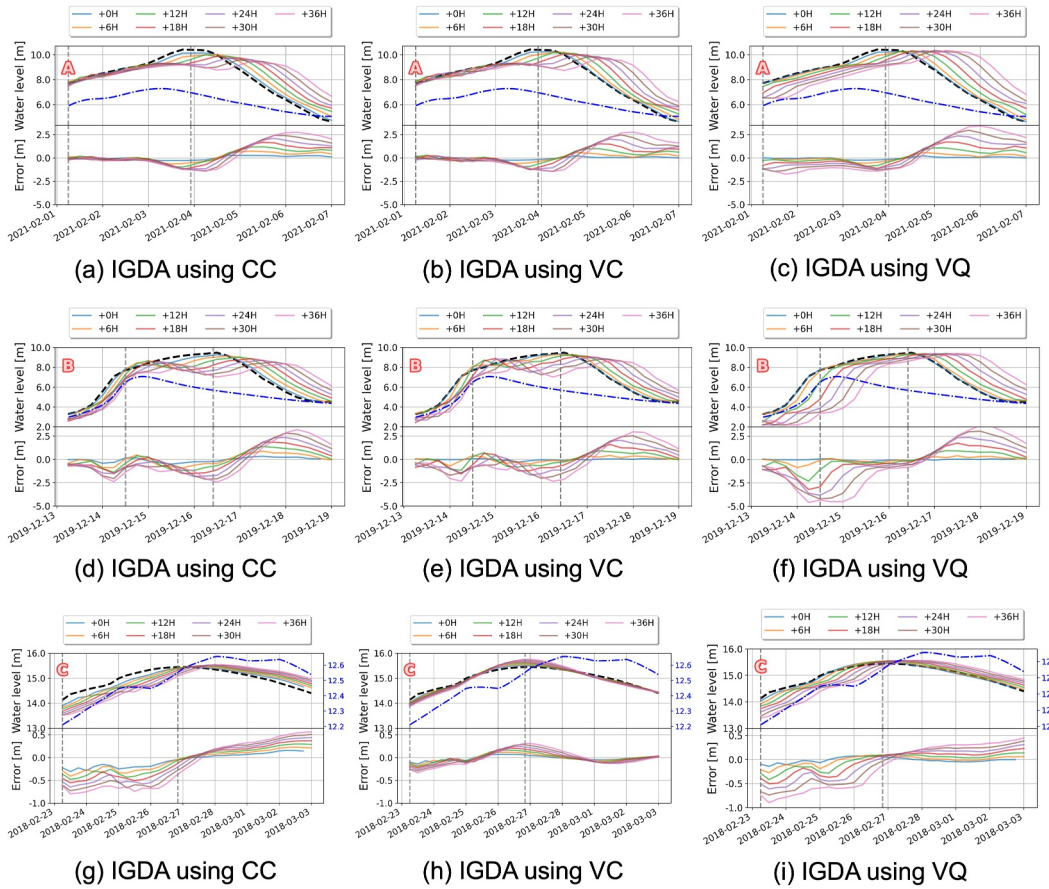


Figure 10. Forecasted water levels at Marmande/Cannelton station (averaged among N_e members) for increasing lead times (+0, +6, +12, +18, +24, +30, and +36 hr) in solid-colored lines (top panel) and their errors (bottom panel) with respect to the observed water levels (black-dashed line). Blue dash-dotted lines show OL^C. Vertical gray dashed lines indicate the threshold exceedance and flood-peak dates. Columns (a-d-g) represent IGDA using CTRIP-CTRIP; (b-e-h) IGDA using VigiCrue-CTRIP; (c-f-i) IGDA using VigiCrue-constant Q. Event A (first row): 2021-02-01 to 2021-02-07; Event B (second row): 2019-12-13 to 2019-12-19; Event C (third row): 2018-02-23 to 2018-03-03.

Results indicate that the forecast succeeds in representing the flood rising limb for all lead times when CTRIP runoff is used as forcing beyond the present time T_0 (i.e., with IGDA using CTRIP-CTRIP in Figure 10a and IGDA using VigiCrue-CTRIP in Figure 10b). On the opposite, prescribing a constant discharge (Figure 10c) past the present time fails to capture the rising limb for lead times beyond 12 hr. As such, large errors ranging from 40 cm (for IGDA using VigiCrue-constant Q +12 hr) to nearly 2 m (for IGDA using VigiCrue-constant Q +36 hr) between 2021-02-01 and 2021-02-03 can be seen on Figure 10c, whereas errors with respect to the observations remain significantly lower for IGDA using CTRIP-CTRIP (Figure 10a) and IGDA using VigiCrue-CTRIP (Figure 10b) for all lead times. It should also be noted that for all three forcing strategies, the flood peak is delayed and underestimated, especially beyond 6 hr lead time. The water level error reaches 3.5 m at Marmande during the flood recess on 2021-02-06 (Figure 10c, for +36 hr). Similar analyses stand for the other events. For instance, for Event B, during the flood rising limb between 2019-12-14 and 2019-12-16, the forecasted water level errors by IGDA using CTRIP-CTRIP (Figure 10d) and IGDA using VigiCrue-CTRIP (Figure 10e) are significanlty lower than those of IGDA VigiCrue-constant Q (Figure 10f). In spite of the limitations regarding flood peak forecasting, the ability to correctly predict the rising limb, in particular when the risk transitioned from a yellow to an orange alert level, is essential for flood monitoring and early warning. The merits of using either VigiCrue forcing (which is not available in real time) or CTRIP forcing over a constant forcing are clearly shown here. Therefore, the forcing strategies using CTRIP-CTRIP and VigiCrue-CTRIP should be favored over VigiCrue-constant Q.

Figure 11 depicts the RMSEs between the in situ observation and IGDA using CTRIP-CTRIP (blue line), VigiCrue-CTRIP (orange line) and VigiCrue-constant Q (green line) forecasts. The RMSEs are computed across

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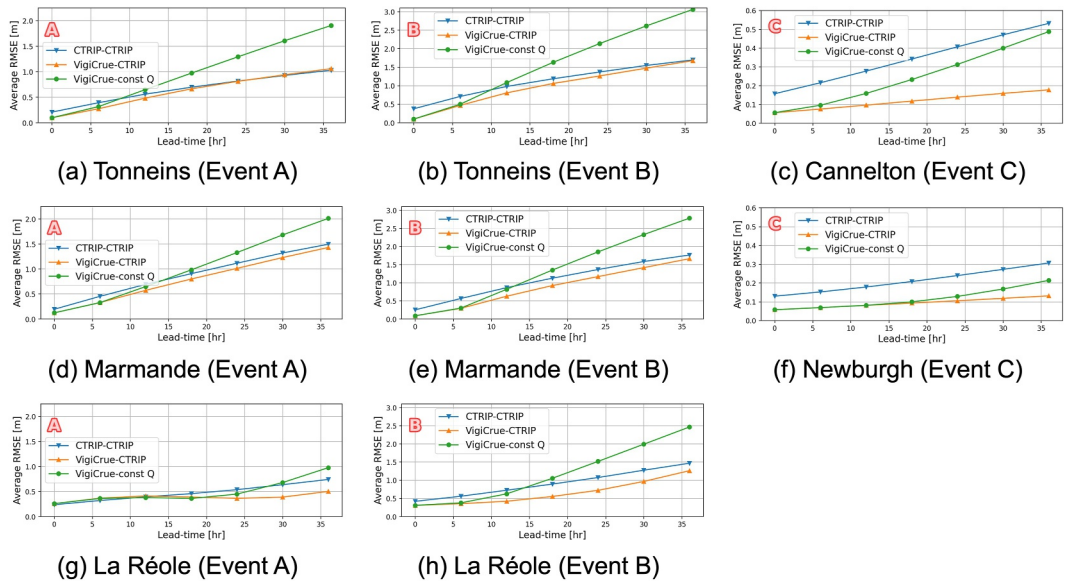


Figure 11. RMSE between forecasted water level (averaged among N_e members) and observation, computed over the flood peak period at Tonneins/Marmande/La Réole (Events A and B) and Cannelton/Newburgh (Event C) along increasing lead time. IGDA using CTRIP-CTRIP (blue lines); IGDA using VigiCrue-CTRIP (orange lines); IGDA using VigiCrue-constant Q (green lines). Event A (first row): 2021-02-01 to 2021-02-07; Event B (second row): 2019-12-13 to 2019-12-19; Event C (third row): 2018-02-23 to 2018-03-03.

6 days (respectively, 8 days) around the flood peak for all observing stations, Tonneins, Marmande, and La Réole (respectively, Cannelton and Newburgh) for Events A and B (respectively, Event C) for increasing lead times. For all forcing strategies, as expected, the forecast performance decreases as the lead time increases, especially at Tonneins, Marmande and Cannelton. The performance of the three strategies does not differ markedly for lead times below 12 hr, albeit with IGDA using CTRIP-CTRIP showing slightly higher RMSEs. Yet, for all three events, using VigiCrue during the reanalysis phase proves most effective, with either CTRIP or a constant discharge during the forecast phase (i.e., orange and blue lines in Figure 11). Consistently, the VC and VQ strategies outperform CC for lead times shorter than 12 hr across all events. It should be noted that the RMSEs for Event C are relatively low, that is, under 50 cm, across the different strategies and stations (Figures 11c and 11f). This stems from the fact that the two stations are actually BCs of the hydraulic model over the Ohio River. For longer lead times, using VigiCrue during the reanalysis phase yields lower RMSEs than using CTRIP. Plus, CTRIP should still be preferred over a constant discharge during the forecast phase. Ultimately, across the three events, the most accurate forecasts for long lead times are obtained with IGDA using VigiCrue-CTRIP.

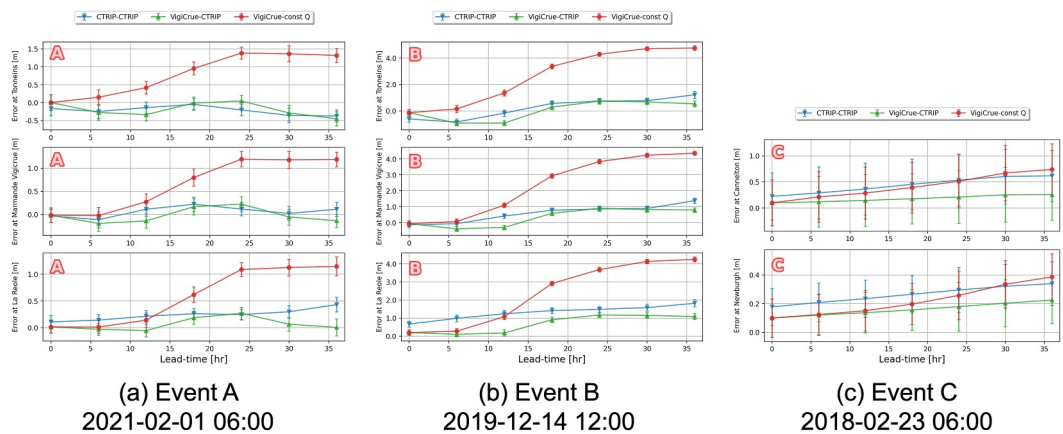


Figure 12. Forecast water level error at Tonneins/Cannelton (top panel), Marmande (middle panel), and La Réole/Newburgh (bottom panel) for the orange threshold exceedance dates during rising limbs 2021-02-01 06:00, 2019-12-14 12:00, and 2018-02-23 06:00 with various lead times for (a) Event A, (b) Event B, and (c) Event C, respectively.

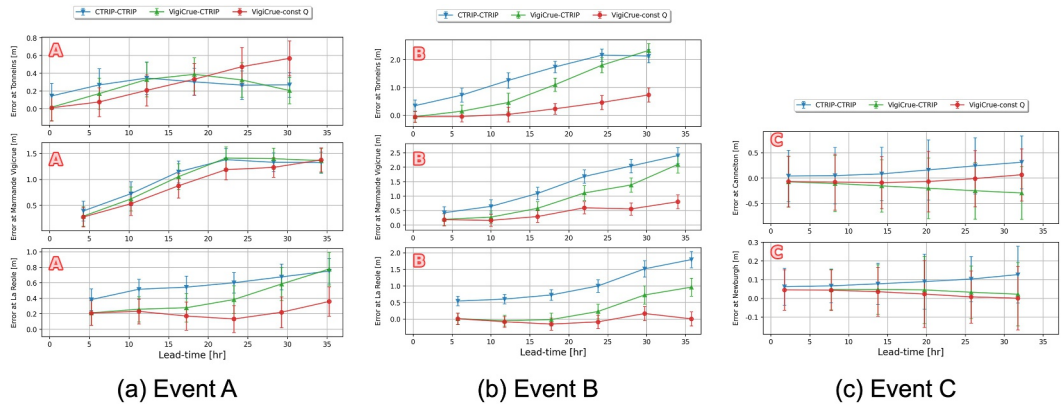


Figure 13. Forecast water level errors at Tonneins/Cannelton (top panel), Marmande (middle panel), and La Réole/Newburgh (bottom panel), at their respective time of peak water level, with various lead times. (a) Event A: 2021-02-03 12:15 (Tonneins), 22:15 (Marmande), 2021-02-04 11:15 (La Réole); (b) Event B: 2019-12-16 00:15 (Tonneins), 10:00 (Marmande), and 18:00 (La Réole); (c) Event C: 2018-02-26 20:15 (Cannelton), and 2018-02-28 01:45 (Newburgh).

Errors in forecasted water levels at observing stations for the orange threshold exceedance dates are displayed in Figure 12, where the observed water depth at Marmande reaches 7.60 m (occurred around 2021-02-01 06:00 for Event A and 2019-12-14 12:00 for Event B) or that of Cannelton to reach 14.02 m (2018-02-23 06:00 for Event C). At this stage, the flood events transitioned from a yellow to an orange alert level where overflows started to occur. The differences between the observed and the forecasted water levels are plotted for Tonneins, Marmande, and La Réole for Events A and B, or for Cannelton and Newburgh for Event C, as a function of increasing lead time varying from +0 hr to +36 hr along *x*-axis. The three forcing strategies—IGDA using CTRIP-CTRIP, using VigiCrue-CTRIP, and using VigiCrue-constant Q—are depicted in blue, green, and red, respectively. Similar results are shown at flood peak dates in Figure 13.

During the rising limbs (Figure 12), the forecast error reaches up to 1.5 m in Event A and 4 m in Event B when constant Q is prescribed (VigiCrue-constant Q) whereas it remains below 1 m when CTRIP forcing is used (CTRIP-CTRIP and VigiCrue-CTRIP). This further reinforces that IGDA using VigiCrue-CTRIP represents the most effective strategy, as previously concluded from Figure 11. Forecast errors for the flood peaks (Figure 13) are substantially larger than for the rising limbs (e.g., 1.5 m at Marmande), as CTRIP struggles to predict peak flows reliably. This is true whether CTRIP is used in the forecast phase following VigiCrue or CTRIP in the reanalysis phase. It should also be noted that, at Tonneins during the flood peak (2021-02-03 12:15), the early peak ISBA-CTRIP discharge shown in Figure 4a, contributes to the slight reduction of the forecast errors beyond 18 hr lead time for CTRIP-CTRIP and VigiCrue-CTRIP. This behavior was not observed for Event B, as CTRIP estimated the flood peak nearly 2 days earlier than the actual peak (Figure 4b), whereas for Event C, CTRIP underestimated the flood peak and produced a broader hydrograph, that is, with a slower rise and recession (Figure 4c).

Figure 14 shows the contingency maps between the forecasted and the observed flood extents for the three forecasting strategies at increasing lead time. Table 4 presents the corresponding CSI scores obtained from the three forecasting strategies with different lead times. With a 12-hr lead time, the forecast method employing IGDA with CTRIP-CTRIP nearly matches the reanalysis CSI score of $IGDA^{C}$ previously shown in Figure 9 (64.68% compared to 68.34%, for Event A). This score is even better for Events B and C (52.84% and 71.21%, compared to 41.33% and 70.87%, respectively). Similarly, there is only a slight disparity between IGDA using VigiCrue-CTRIP and $IGDA^{V}$ (68.19% vs. 68.96%) in Event A. The same remark applies for Event B and C (52.66% and 71.93%, vs. 51.16% and 71.32%, respectively).

As expected, longer lead times generally result in lower CSI scores. Indeed, in Event A, a notable degradation between +12 hr and +30 hr (69.67% and 54.32%), occurs in the third approach, that is, IGDA using VigiCrue-constant Q. This reemphasizes the large underestimations of the discharge for the VigiCrue-constant Q strategy (in the part leading up to the flood peak and the peak itself) when increasing lead times with this approach, as previously shown in Figure 10c. At 30-hr lead time, the CSI yields 54.32% when using a constant forecast (IGDA with VigiCrue-constant Q), whereas it increases to 63% when employing CTRIP-based predictions (IGDA with

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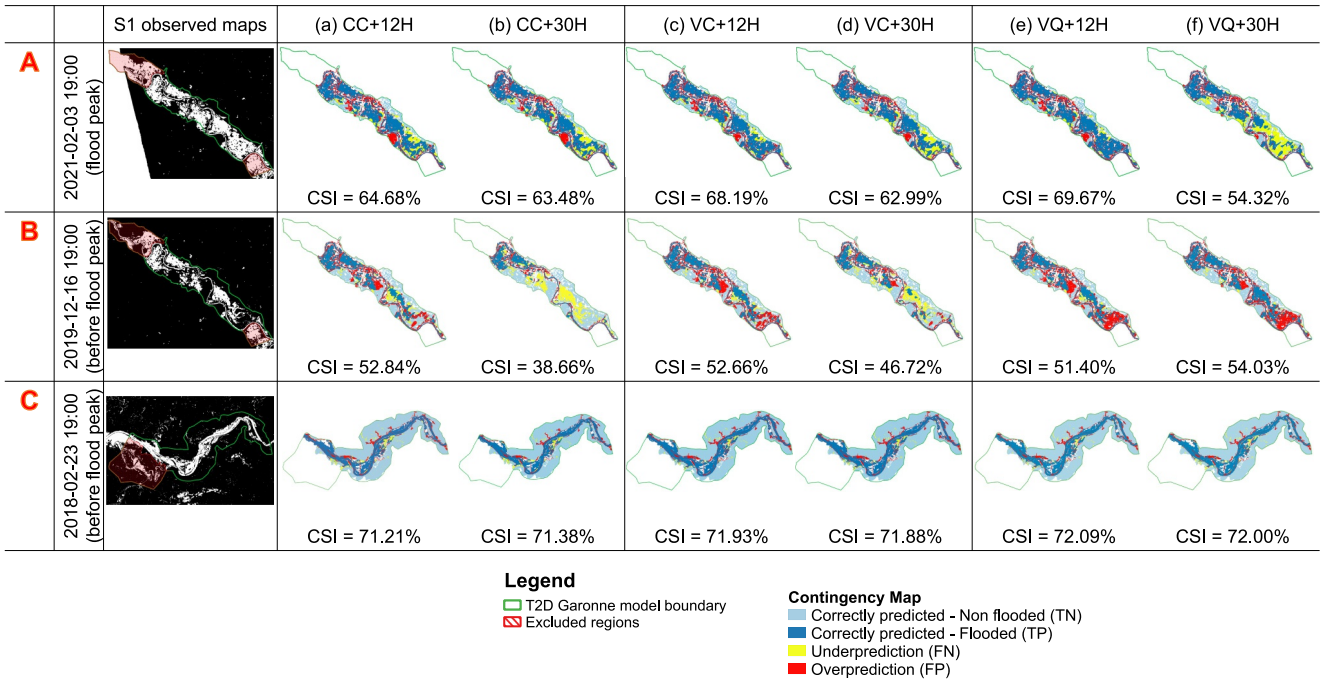


Figure 14. Contingency maps and Critical Success Index computed between forecasted flood extents (averaged among N_e members) and S1-derived observed flood extents for IGDA using the three different approaches at the flood peak for Event A (first row), before flood peak for Event B (second row) and C (third row). Panels (a) and (b): IGDA using CTRIP-CTRIP with lead time +12 hr and +30 hr; Panels (c) and (d): IGDA using VigiCrue-CTRIP with lead time +12 hr and +30 hr; Panels (e) and (f): IGDA using VigiCrue-constant Q with lead time +12 hr and +30 hr.

VigiCrue-CTRIP and CTRIP-CTRIP). This indicates that CTRIP forcing substantially improves the accuracy of flood extent predictions compared to Sentinel-1 imagery during Event A. However, this improvement is not observed for Events B and C, where CTRIP forcing at a 30-hr lead time does not provide additional benefit, primarily due to the underestimated magnitude, timing lag, and prolonged peak in the CTRIP discharge. It is noteworthy that the observed degradation primarily occurs in the upstream region of the catchment, which is heavily influenced by the inflow dynamics. It should also be noted that, for Event C, all CSI values are nearly identical, with only a 1.02-percentage-point difference between the lowest and highest values. Overall, the IGDA using the VigiCrue–CTRIP configuration is the most stable among the three forecast strategies across the three events.

4.4. Post-Event Assessment With High Water Marks Observations

The results are also evaluated using independent observations from relevant HWM data set. Such a collaborative data set consists of flood marks (https://www.reperesdecrues.developpement-durable.gouv.fr/) collected and validated by the local flood forecast services (SPCs) and/or "Établissements Publics Territoriaux de Bassins" (EPTB) (Fromental et al., 2024). In the aftermath of the 2021 flood event (Event A), 178 HWM observations were collected; They were accessed in December 2022. This assessment is only carried out for Event A, which was more significant than Event B. To the best of our knowledge, HWM observations are not available for the Ohio catchment during Event C.

A validation of the performed DA strategies and the three forecast approaches (focusing on IGDA) with respect to independent data was finally carried out using the collective public data sets of HWM for the Event A. This allows the evaluation of the highest simulated water levels spatially distributed at various points on the riverbanks and within the floodplain at the flood peak. Figure 15 presents the validation results based on HWMs for the flood event. Negative (respectively, positive) discrepancies between maximum simulated/forecasted water levels and observed HWM are indicated with yellow (respectively, red) triangles of varying size. Figure 15a depicts the comparison between the OL using a persistent discharge for forecast (OL VQ), IGDA using VigiCrue-constant Q (IGDA VQ) and IGDA using VigiCrue-CTRIP (IGDA VC), for forecast lead time +30 hr Figure 15b

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Table 4
Critical Success Index Between Forecasted Flood Extent Maps and S1-Derived Observed Flood Extent Maps

	CSI [%] at 2021-02-03 19:00 (flood peak)		
Forecast (hr)	IGDA using CC (%)	IGDA using VC (%)	IGDA using VQ (%)
(a) Event A–Garonne (2021)			
+6 hr	67.03	69.76	**70.08**
+12 hr	64.68	68.19	**69.67**
+18 hr	61.11	64.91	**68.14**
+24 hr	62.17	**66.54**	62.65
+30 hr	**63.48**	62.99	54.32
	CSI [%] 2019-12-16 19:00 (before flood peak)		
Forecast (hr)	IGDA using CC (%)	IGDA using VC (%)	IGDA using VQ (%)
(b) Event B - Garonne (2019)			
+6 hr	**51.76**	50.89	50.86
+12 hr	**52.84**	52.66	51.40
+18 hr	47.58	**54.32**	51.94
+24 hr	44.94	**56.17**	52.67
+30 hr	38.66	46.72	**54.03**
	CSI [%] at 2018-02-23 19:00 (before flood peak)		
Forecast (hr)	IGDA using CC (%)	IGDA using VC (%)	IGDA using VQ (%)
(c) Event C - Ohio (2018)			
+6 hr	70.98	71.72	**71.77**
+12 hr	71.21	71.93	**72.09**
+18 hr	71.08	71.64	**71.79**
+24 hr	71.16	71.76	**71.92**
+30 hr	71.38	71.88	**72.00**

Note. Highest CSIs are boldfaced.

(respectively, Figure 15c) shows the scatter plot comparing simulated/forecasted highest water levels (respectively, errors with respect to the observed HWMs) on *x*-axis versus observed HWMs on *y*-axis.

The DA strategy, employing a coupled hydrological-hydraulic modeling framework (IGDA VC, right panel in Figure 15a), demonstrates clear advantages over the baseline operational approach: an OL simulation without DA using observed discharge up to the present and keep it throughout the forecast period (Figure 15a, left panel). Notably, the discrepancy between the forecasted maximum water levels and the observed HWMs is significantly reduced across the entire Garonne catchment, with particularly marked improvements in the Marmande area (see zoomed view). When VigiCrue forcing is used over the reanalysis phase along with DA corrections (Figure 15a, middle and right panels), the use of CTRIP forcing (IGDA VC) in place of persistent inflow (IGDA VQ) results in improved forecast in the upstream part of the catchment for +30 hr lead time.

Table 5 summarizes the RMSE of the performed DA experiments and forecast approaches, between the simulated/forecast maximum water levels and the 178 HWM observations, in Event A. In reanalysis, DA decreases the RSME to independent HWM observations when either VigiCrue or CTRIP forcing is used. This outcome was anticipated, as both OL^V and OL^C were found to underestimate water levels at several gauging stations, suggesting a broader underestimation across the entire simulation domain. In forecast mode (at lead times under 12 hr), the RMSEs stay near those of the reanalysis, ranging between 2 and 10 cm, indicating that the real-time forecasting workflow is effective and can attain high accuracy. Among the three strategies, IGDA using VigiCrue-constant Q produces the lowest RMSE for lead times under 18 hr. However, for longer lead times, the forecast approach using IGDA with CTRIP-CTRIP performs slightly better in predicting the maximum water levels over the entire flood event.

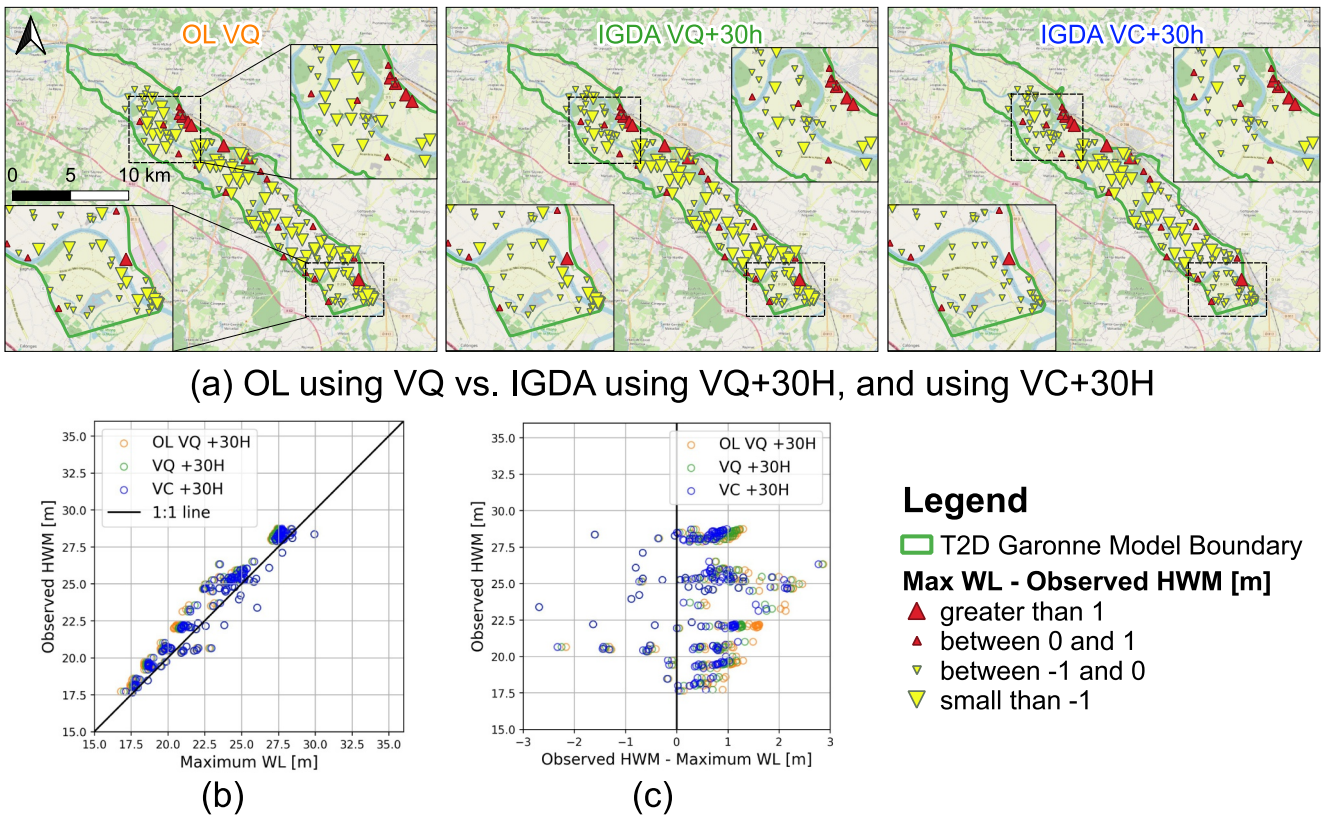


Figure 15. (a) Post-event High Water Marks validation for Event A between OL using VigiCrue-constant Q and IGDA using VigiCrue-CTRIP, and IGDA using VigiCrue-constant Q, and using VigiCrue-CTRIP with lead time +30 hr. Negative values (yellow triangles) indicate underpredictions by the simulations, while positive values (red triangles) indicate overpredictions. (b) Maximum water levels against observed HWMs with a 1:1 line for clarity. (c) Van de Casteele diagram, showing errors against observed HWMs.

Lastly, Table 6 summarizes the performance of the three forecasting strategies (CC, VC, and VQ), against each other, in the observation space across the three study events. It consolidates results from various assessments, that is, water level RMSEs at gauging stations, CSI scores of flood extent and HWM RMSEs post event. It highlights that the VC configuration offers the most consistent forecast accuracy across all metrics and lead times, while CC tends to perform better at extended lead times.

Table 5
RMSE of Simulated/Forecasted Maximum Water Levels and High Water Marks Observations for the Event A

		RMSE [m]	
Reanalysis	OL V	IDA V	IGDA V
	1.237	**1.047**	1.064
	OL C	IDA C	IGDA C
	1.695	1.077	**1.068**
Forecast	IGDA CC	IGDA VC	IGDA VQ
+6 hr	1.110	1.077	**1.022**
+12 hr	1.162	1.146	**1.085**
+18 hr	1.210	1.226	**1.175**
+24 hr	**1.180**	1.226	1.227
+30 hr	**1.171**	1.203	1.278

Note. Lowest RMSEs are boldfaced.

5. Conclusion and Perspectives

Reliable real-time forecasts over extended periods are crucial for predicting and preventing flood risks, enabling timely and cost-effective mitigation measures. The present research work aims at such an operational flood forecasting chain. It integrates runoff predictions from an ISBA-CTRIP large-scale hydrologic model as inputs into a high-resolution, high-fidelity, local T2D hydrodynamic model to simulate and forecast water levels and flood extents. Such an integration was demonstrated over two study catchments of the Garonne and Ohio Rivers, over three major flood events in 2021, 2019, and 2018, referred to as Events A, B, and C, respectively. This paper highlights the merits of jointly assimilating in situ water-level observations and 2D flood extent maps derived from Sentinel-1 SAR imagery using an EnKF implemented in T2D models, in order to address the substantial uncertainties in hydrological forcings and hydraulic model parameters in an operational flood forecasting context. The EnKF control vector includes friction

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Table 6
Comparison of the Three Forecast Strategies, Ranked From I (Best) to III (Worst)

Assessment	Involved events	Illustration	Forecast strategy		
			IGDA CC	IGDA VC	IGDA VQ
Mean WL (<18 hr)	A, B, C	Figure 11	III	I	II
Mean WL (≥18 hr)	A, B, C	Figure 11	II	I	III
WL at rising limb	A, B, C	Figure 12	II/III	I	II/III
WL at peak	A, B, C	Figure 13	III	II	I
Contingency maps (≤18 hr)	A, B, C	Table 4	III	II	I
Contingency maps (+24 hr)	A, B, C	Table 4	III	I	II
HWM (≤18 hr)	A	Table 5	III	II	I
HWM (>18 hr)	A	Table 5	I	II	III

coefficients and forcing correction and is augmented with correction of the hydraulic state in subdomains of the floodplains.

It was shown that the integration of advanced technologies such as machine learning, remote sensing, and hydrological and hydraulic modeling has enhanced the accuracy and reliability of flood forecasting systems. These tools allow for more precise, more adaptive and real-time responses, improving the resilience of communities and infrastructure against climate-related threats. In addition, emphasizing local-scale predictions that account for unique geographic and climatic conditions can enhance the effectiveness of flood risk management strategies. Further research works will also focus on incorporating climate change projections into forecasting models to better predict future flood scenarios.

The DA framework is cycled with a real-time forecasting configuration. A cycle consists of a reanalysis phase and a forecast phase. Over the 6-hr reanalysis, observations up to the present are assimilated. An ensemble is then initialized from the last reanalyzed states and issued forecasts for the next 36 hr. In the reanalysis mode, six hindcast experiments were carried out both in Open Loop and DA, using upstream forcing data that are either observed discharges from an observing station in the VigiCrue (Events A and B) or USGS (Event C) network, or discharges simulated by a large-scale hydrologic model ISBA-CTRIP. In the real-time forecast mode, three strategies of forcing data for this forecast are performed following IGDA reanalysis: (a) using CTRIP predicted runoff for analysis and forecast, (b) using observed discharge for analysis and then CTRIP runoff for forecast and (c) using observed discharge for analysis and keep a persistent discharge value for forecast.

In hindcast mode, IGDA was shown to substantially improve water level predictions in the riverbed and the representation of flow dynamics within the floodplain. This improvement is observed when using either discharge observations from VigiCrue or large-scale hydrologic runoff from ISBA-CTRIP. The effectiveness of DA was evaluated using both 1D and 2D metrics. While assimilating in situ data corrects the hydraulic state in the riverbed, the assimilation of WSR further enhances the representation of flow dynamics in the floodplain. Indeed, DA successfully identifies corrections to friction coefficients, hydrologic forcings, and the hydraulic state in the floodplain, ensuring that the simulated hydraulic conditions are consistent with both in situ and remote-sensing observations. This highlights the advantages of jointly assimilating in situ and remote-sensing data for reanalyzing past flood events, resulting in a more accurate hydraulic state for initializing ensemble forecasts.

In forecast mode, combining observed forcing during the reanalysis phase with CTRIP-predicted forcing for the forecast phase (i.e., VigiCrue–CTRIP) yields the most reliable water level and flood extent predictions. For all three strategies considered, forecast quality decreases with increasing lead time. When CTRIP runoffs are used during the forecast phase, employing observed discharge (i.e., VigiCrue-CTRIP) or CTRIP runoff (i.e., CTRIP-CTRIP) as forcing during the reanalysis phase yields comparable performance. This approach proves superior to using a persistent Q value for forecast forcing (i.e. VigiCrue-constant Q), even though the quality of the CTRIP predicted runoff (in terms of amplitude and phase) may influence the conclusions at specific dates over different flood events. In the present study, the gain of the chained hydrologic-hydrodynamic models was all the more visible during the rising limbs of the floods. The 2D evaluation of the chained DA strategy using remote sensing data provides additional support for incorporating hydrological predictions into the forecasting framework. This

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approach yields more accurate representations of the flood extents. This is further supported by the comparison between the chained forecast results and observed HWMs when available.

This study demonstrates that, despite some imperfections, forcing data from a large-scale hydrologic model can be effectively used as input to a local hydraulic model with DA for both reanalysis and real-time forecasting. The results support a multi-source assimilation strategy for the algorithm applied to a chained hydrologic-hydraulic model, proving beneficial for short-term forecasts and extended lead times, particularly in poorly gauged catchments.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

- The in situ hydrometric data used in this study are available via the French national hydrometric data portal: https://hydro.eaufrance.fr/.
- Sentinel-1 satellite data provided under Copernicus Service by the European Union and ESA. They are available to download from NASA's Alaska Satellite Facility Distributed Active Archive Center (ASF DAAC). FloodML codes that process Sentinel-1 SAR images are available at Rodriguez Suquet (2025).
- TELEMAC-2D was originally developed by the National Hydraulics and Environment Laboratory (Laboratoire National d'Hydraulique et Environnement – LNHE) within the EDF's R&D division. It is now maintained by a consortium of consulting firms and research institutions, including CERFACS. TELEMAC-MASCARET, the broader integrated suite of solvers for free-surface flow modeling, is distributed under the GNU General Public License, version 3. Further information can be found on the software official website (https://www.opentelemac.org).
- The CERFACS-developed data assimilation (DA) algorithms can be found at https://gitlab.pam-retd.fr/otm/telemac-mascaret/-/tree/fishstick. The DA framework is designed with operational efficiency in mind to meet the demands of real-time forecasting. On average, the DA workflow requires approximately 8–10 min per reanalysis cycle (combining one gray, one blue and one green box from Figure 5) and approximately 19–20 min per cycle (combining orange and red boxes from Figure 5) to issue a forecast for the next 30 hr. These benchmarks were obtained on a high-performance computing system at CERFACS, utilizing 25 dual-socket compute nodes. Each node is equipped with 18-core Intel® Xeon® Gold 6140 processors (Skylake, 2.3 GHz) and 96 GB of RAM, with parallelization managed via OpenMPI.

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